Output Reference

SWAT+ Documentation

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SWAT+ writes text files with results for each object type (HRU, landscape unit, channel, aquifer, reservoir and so on) and for the basin as a whole, by day, month, year and as an average over the simulation. Which files are written is set in `print.prt`: each row names a group of outputs and has a `y` or `n` for the daily, monthly, yearly and average annual files. A few files are written in every run.

The files go to the output folder named in `file.cio`, or to the run folder when none is named. Every run also writes `files_out.out`, a list of the output files it opened, which is the quickest way to see what a given `print.prt` produced.

Pages

Page	Content
Naming convention	How files are named and laid out, print timing, and the `print.prt` row behind every file.
Water balance	`hru_wb`, `hru-lte_wb`, `lsunit_wb` and `basin_wb`: precipitation, evapotranspiration, runoff, percolation, soil water and snow.
Channel, sediment, nutrient	`channel_sd`, `channel_sdmorph`, `sd_chanbud`, the sediment budget files `ch_sedbud`, `chanbud_order` and `sedbud`, their basin sums, and the legacy `channel` files.
Hydrographs	`hydout`, `hydin`, `deposition` and `ru`: water and loads passed between objects, and the `hydcon.out` connection list.
Recall	`recall` and `basin_psc`: loads released by point sources, inlets and diversions.
Nutrient balance and losses	`hru_nb`, `hru_ls`, `lsunit_nb`, `lsunit_ls`, `basin_nb`, `basin_ls` and `hru-lte_ls`: nitrogen and phosphorus balances and the sediment and nutrient losses from the landscape.
Aquifer, reservoir, plant	`aquifer`, `reservoir`, `wetland`, crop yields and the plant and weather (`_pw`) files.
Carbon	Soil, residue and plant carbon files.
gwflow	Outputs of the gridded groundwater module.
Checker and diagnostics	Files written in every run: `checker.out`, `diagnostics.out`, `files_out.out` and others.
Post-processing	The Python scripts in the `test` folder of the SWAT+ source.

Recent changes

- 62.0.1 adds the channel sediment budget files `ch_sedbud_*` (each channel), `chanbud_order_*` (each stream order) and `sedbud_*` (basin), all switched on by the `channel_sd` row.
- 62.0.0 added the carbon rows of `print.prt` (`hru_cb_gl`, `hru_cb_trf`, `lsu_cb_gl` and others) for the carbon output families; see Carbon.

  Sources

  - `print.prt` reader: `basin_print_codes_read.f90`
  - Output folder: `output_path_module.f90`
  - File opening: `proc_open.f90`, `output_landscape_init.f90`, `header_write.f90`, `header_sd_channel.f90`
  - `files_out.out`: `proc_bsn.f90`

Filename pattern

Most time-step outputs are named

`<object>_<group>_<interval>.txt`

for example `hru_wb_day.txt` (HRU water balance, daily) or `basin_sd_cha_aa.txt` (basin sum of channel outputs, average annual). Some files have no group (`aquifer_day.txt`, `reservoir_mon.txt`, `ru_yr.txt`), and a few files written once per run have no interval (`hru_cbn_lyr.txt`, `mgt_out.txt`).

Interval	Meaning	`print.prt` column
`_day`	one row per object per day	`daily`
`_mon`	one row per object per month	`monthly`
`_yr`	one row per object per year	`yearly`
`_aa`	one row per object, average annual over the printed years	`avann`

The channel sediment budget files `ch_sedbud_*` and `chanbud_order_*` use `_d`, `_m`, `_y` and `_a` instead.

When `csvout` is `y` in `print.prt`, most time-step files also get a comma-separated copy with the same name and the extension `.csv`.

Output folder

Output files are written to the folder named on the output path line at the end of `file.cio`. SWAT+ creates the folder if it does not exist. When the line is missing, empty or `null`, outputs go to the folder SWAT+ runs in. `simulation.out`, `success.fin` and `erosion.txt` always go to the run folder.

File layout

Time-step files share one layout:

- Line 1: the project name from `file.cio` and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units (blank for columns without units).
- Lines 4 and on: data rows.

Rows with one object per row start with `jday`, `mon`, `day`, `yr`, `unit`, `gis_id` and `name`. Basin files use the same columns with `unit` and `gis_id` set to 1. `jday`, `mon`, `day` and `yr` are the date of the last day in the print step.

Print timing

The first blocks of `print.prt` set when rows are written:

- Daily rows are written from `day_start` of `yrc_start` to `day_end` of `yrc_end`, every `interval` days. A zero `day_start` means day 1, a zero `day_end` means day 366, and zero years mean the first and last simulated year.
- Monthly, yearly and average annual rows start after the first `nyskip` simulated years. Average annual values are averages over the remaining years.
- The average annual print intervals (`aa_int_cnt` and the end years that follow it) are read, but in 62.0.1 they are used only by the region output, which is not written, so they have no effect.

print.prt rows and files

Each row in the objects block of `print.prt` has a name and four `y`/`n` switches. When `use_obj_lbls` is `y`, rows are matched by name, may come in any order, and rows can be left out (they default to `n`). A name that appears twice stops the run with an error, and so does a name SWAT+ does not know. When `use_obj_lbls` is `n`, the rows are read by position and must follow the order of the table below up to `wetland_cs`; the carbon and gwflow rows can only be set in labelled mode.

Each `y` opens the file for that interval. The table shows the file stems; add `_day.txt`, `_mon.txt`, `_yr.txt` or `_aa.txt`.

`print.prt` row	Files	Described in
`basin_wb`	`basin_wb`	Water balance
`basin_nb`	`basin_nb`	Nutrient balance and losses
`basin_ls`	`basin_ls`	Nutrient balance and losses
`basin_pw`	`basin_pw`	Aquifer, reservoir, plant
`basin_aqu`	`basin_aqu`	Aquifer, reservoir, plant
`basin_res`	`basin_res`	Aquifer, reservoir, plant
`basin_cha`	`basin_cha` (legacy channels)	Channel and sediment
`basin_sd_cha`	`basin_sd_cha`, `basin_sd_chamorph`, `basin_sd_chanbud`	Channel and sediment
`basin_psc`	`basin_psc` (recall objects summed)	Recall
`region_wb`, `region_nb`, `region_ls`, `region_pw`, `region_aqu`, `region_res`, `region_sd_cha`, `region_psc`	none in 62.0.1	
`water_allo`	`water_allo`, `wallo_use`, `wallo_treat`	Water allocation

print.prt row	Files	Described in
lsunit_wb	lsunit_wb	Water balance
lsunit_nb, lsunit_ls, lsunit_pw	lsunit_nb, lsunit_ls, lsunit_pw	Nutrient balance and losses, Aquifer, reservoir, plant
hru_wb	hru_wb; also hru_path when constituents are simulated	Water balance
hru_nb	hru_nb, hru_ncycle	Nutrient balance and losses
hru_ls	hru_ls	Nutrient balance and losses
hru_pw	hru_pw	Aquifer, reservoir, plant
hru-lte_wb, hru-lte_nb, hru-lte_ls, hru-lte_pw	hru-lte_wb, hru-lte_ls, hru-lte_pw	Water balance, Nutrient balance and losses, Aquifer, reservoir, plant
channel	channel (legacy channels)	Channel and sediment
channel_sd	channel_sd, channel_sdmorph, sd_chanbud, sedbud, ch_sedbud, chanbud_order, channel_sd_subday.txt	Channel and sediment
aquifer	aquifer	Aquifer, reservoir, plant
reservoir	reservoir, wetland	Aquifer, reservoir, plant
recall	recall	Recall
hyd	hydin, hydout, deposition	Hydrographs
ru	ru (routing unit hydrographs)	Hydrographs
pest	hru_pest, channel_pest, reservoir_pest, aquifer_pest, basin_ls_pest, basin_ch_pest, basin_res_pest, basin_aqu_pest	
basin_salt, hru_salt, ru_salt, aqu_salt, channel_salt, res_salt, wetland_salt	basin_salt, hru_salt, rout_unit_salt, aquifer_salt, channel_salt, reservoir_salt, wetland_salt	Salts
basin_cs, hru_cs, ru_cs, aqu_cs, channel_cs, res_cs	basin_cs, hru_cs, rout_unit_cs, aquifer_cs, channel_cs; res_cs opens both reservoir_cs and wetland_cs	Constituents
wetland_cs	none (wetland constituent files follow res_cs)	
hru_cb, hru_cb_vars, hru_cb_gl, hru_cb_trf, hru_cb_lyr, hru_cb_cpool, hru_cb_npool, hru_cb_plt, hru_cb_flux, hru_cb_drv, hru_cb_dyn, hru_cb_snap, lsu_cb_gl, lsu_cb_trf, lsu_cb_plt	carbon files	Carbon
gwflow_wb, gwflow_flux, gwflow_heat, gwflow_solute, gwflow_obs, gwflow_pump	gwflow files	gwflow

The rows hru-lte_nb and region_* are read but open no files in 62.0.1.

Other switches

The line after csvout in print.prt has four more switches:

Switch	Values	Files
crop_yld	a (average annual), y (yearly), b (both), n	crop_yld_aa.txt, crop_yld_yr.txt, basin_crop_yld_aa.txt, basin_crop_yld_yr.txt; y and b also open yield.out
mgtout	y, n	mgt_out.txt, the log of management operations
hydcon	y, n	hydcon.out, the object connection list (see Hydrographs)
fdcout	y, n	flow_duration_curve.out

cdfout (netCDF output) is read but not used.

Files written every run

File	Content
files_out.out	List of every output file opened in the run, with a short tag.
diagnostics.out	Warnings about missing or unmatched inputs.
checker.out	Soil and HRU parameter summary.
erosion.out	USLE factors per HRU.
area_calc.out	Area checks for routing units and landscape units.
lu_change_out.txt	Land use and parameter changes made by decision tables.
simulation.out	Run log.
success.fin	Written when the run finishes normally.

See Checker and diagnostics for their contents.

Related files

- print.prt: the switches described on this page.
- file.cio: names the output folder.
- Output reference: overview of the output pages.

Sources

- print.prt reader: basin_print_codes_read.f90; switch type basin_print_codes in basin_module.f90
- Output folder: output_path_module.f90, read in readcio_read.f90
- Header writers called at startup: proc_open.f90, which calls output_landscape_init.f90, header_write.f90, header_sd_channel.f90 and the other header_*.f90 routines
- Files written every run: proc_bsn.f90, proc_hru.f90, main.f90.in
- Print window: time_control.f90

print.prt item	Fortran variable
nyskip, day_start, yrc_start, day_end, yrc_end, interval	pco%nyskip, pco%day_start, pco%yrc_start, pco%day_end, pco%yrc_end, pco%int_day
aa_int_cnt and end years	pco%aa_numint, pco%aa_yrs
csvout, use_obj_lbls, cdfout	pco%csvout, pco%use_obj_labels, pco%cdfout
crop_yld, mgtout, hydcon, fdcout	pco%crop_yld, pco%mgtout, pco%hydcon, pco%fdcout
object row <name>	pco%<group>%d, %m, %y, %a (for example hru_wb -> pco%wb_hru, channel_sd -> pco%sd_chan, basin_psc -> pco%recall_bsn)

Purpose

The water balance files report the daily water budget of the land surface: precipitation and snow, evapotranspiration, surface runoff, lateral and tile flow, percolation, soil water and snowpack storage, and where the runoff goes. Every flux is a depth of water in mm over the area of the object, so objects of different size can be compared directly.

Four scopes are written, each switched on by a row in `print.prt`:

`print.prt` row	Files	One row per
`hru_wb`	`hru_wb_day.txt`, `hru_wb_mon.txt`, `hru_wb_yr.txt`, `hru_wb_aa.txt`	HRU
`hru-lte_wb`	`hru-lte_wb_day.txt`, `hru-lte_wb_mon.txt`, `hru-lte_wb_yr.txt`, `hru-lte_wb_aa.txt`	HRU-LTE
`lsunit_wb`	`lsunit_wb_day.txt`, `lsunit_wb_mon.txt`, `lsunit_wb_yr.txt`, `lsunit_wb_aa.txt`	landscape unit
`basin_wb`	`basin_wb_day.txt`, `basin_wb_mon.txt`, `basin_wb_yr.txt`, `basin_wb_aa.txt`	the whole basin (one row)

Each `y` in the `daily`, `monthly`, `yearly` or `avann` column of a row opens the matching file. With `csvout` set to `y`, a comma-separated `.csv` copy of each file is written as well.

`print.prt` also accepts a `region_wb` row, but 62.0.1 writes no region files, so the row has no effect. Routing units have no water balance file: the `ru` row writes routing unit hydrographs (`ru_day.txt` and so on; see Hydrograph outputs).

Format

- Line 1: the project name from `file.cio` and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units.
- Lines 4 and on: one row per object per print step.

The first seven columns are the same in every landscape output file:

#	Column	Description
1	`jday`	Day of the year of the last day in the print step (for example 31 for a January row of a monthly file).
2	`mon`	Month.
3	`day`	Day of the month.
4	`yr`	Calendar year.
5	`unit`	Sequence number of the object (HRU number, landscape unit number). The basin file always shows 1.
6	`gis_id`	GIS identifier of the object from its connection file. The basin file always shows 1.
7	`name`	Object name. The basin file shows the project name.

The remaining columns, in file order:

#	Column	Units	Description
8	`precip`	mm	Precipitation, rain and snow, before canopy interception.
9	`snofall`	mm	Precipitation that fell as snow.
10	`snomlt`	mm	Snowmelt.
11	`surq_gen`	mm	Surface runoff that leaves the HRU on the day, after the surface runoff lag.
12	`latq`	mm	Lateral flow out of the soil profile.
13	`wateryld`	mm	Water yield: `surq_gen` + `latq` + `qtile`.
14	`perc`	mm	Percolation out of the bottom of the soil profile.
15	`et`	mm	Actual evapotranspiration: `ecanopy` + `eplant` + `esoil`.
16	`ecanopy`	mm	Evaporation of water intercepted by the canopy.
17	`eplant`	mm	Plant transpiration.
18	`esoil`	mm	Soil evaporation, including evaporation of ponded water.
19	`surq_cont`	mm	Surface runoff generated on the HRU on the day, before the surface runoff lag. For an HRU with a wetland, this is the wetland outflow.
20	`cn`	-	Curve number used for the day.
21	`sw_init`	mm	Soil water content at the start of the print step.
22	`sw_final`	mm	Soil water content at the end of the print step.
23	`sw_ave`	mm	Average soil water content over the print step.
24	`sw_300`	mm	Soil water content of the top 300 mm of soil.
25	`sno_init`	mm	Snow water content at the start of the print step.
26	`sno_final`	mm	Snow water content at the end of the print step.
27	`snopack`	mm	Average snow water content over the print step.
28	`pet`	mm	Potential evapotranspiration.
29	`qtile`	mm	Tile drain flow.
30	`irr`	mm	Irrigation water applied.
31	`surq_runon`	mm	Surface runoff received from upslope objects.
32	`latq_runon`	mm	Lateral flow received from upslope objects.
33	`overbank`	mm	Overbank flood water received from channels.
34	`surq_cha`	mm	Surface runoff sent to channels.
35	`surq_res`	mm	Surface runoff sent to reservoirs.
36	`surq_ls`	mm	Surface runoff sent to other landscape objects.
37	`latq_cha`	mm	Lateral flow sent to channels.
38	`latq_res`	mm	Lateral flow sent to reservoirs.
39	`latq_ls`	mm	Lateral flow sent to other landscape objects.
40	`gwsoilq`	mm	Groundwater moved into the soil profile when the water table rises into it (gwflow only).
41	`satex`	mm	Saturation excess flow from a high water table (gwflow only).
42	`satex_chan`	mm	Saturation excess flow that reaches a channel (gwflow only).
43	`sw_change`	mm	Change in soil water content over the day.
44	`lagsurf`	mm	Surface runoff held back by the lag and still on its way to the channel.
45	`laglatq`	mm	Lateral flow held back by the lag and still on its way to the channel.
46	`lagsatex`	mm	Saturation excess flow held back by the lag and still on its way to the channel.
47	`wet_evap`	mm	Evaporation from the wetland water surface.
48	`wet_oflo`	mm	Wetland outflow (spill).
49	`wet_stor`	mm	Water stored in the wetland.
50	`plant_cov`	text	Plant community of the HRU's land use, from `landuse.lum`.
51	`mgt_ops`	text	Management schedule of the HRU's land use, from `landuse.lum`.

The columns `surq_cha` to `latq_ls` split the HRU's generated surface runoff and lateral flow by receiving object, using the fractions in the HRU's connection file. They are filled for HRUs only and stay zero in HRU-LTE rows.

Trailing columns

The header line lists `plant_cov` and `mgt_ops` in every file, but only the `hru_wb` files write them. The other files end after `wet_stor`, except `basin_wb_aa.txt`, which ends with the soft calibration label and the adjustment value: `Original Simulation` and `0.000` when no soft calibration is run. The label is two words, so a tool that splits on spaces finds one more value than the header has names.

Aggregation

- Daily rows are written for days inside the `print.prt` daily window (`day_start` of `yrc_start` to `day_end` of `yrc_end`), every `interval` days.
- Monthly, yearly and average annual rows start after the `nyskip` warm-up years.
- Fluxes (every column except the storages listed below) are summed over the month or year. The average annual value is the sum over the printed years divided by the number of printed years.
- `cn`, `sw_ave`, `sw_300`, `snopack` and `wet_stor` are averaged over the days in the step. In the average annual file they are averaged over all printed days.
- `sw_init` and `sno_init` give the value at the start of the step, and `sw_final` and `sno_final` the value at the end. In `*_aa.txt`, `sw_final` and `sno_final` are the values on the last day of the simulation.

Landscape unit and basin rows are area-weighted averages of their HRUs and HRU-LTEs, using the area fractions in `ls_unit.ele`.

Example

The header line of `hru_wb_aa.txt` as 62.0.1 writes it, split here over several lines:

```
 jday    mon    day     yr    unit  gis_id  name            precip     snofall      snomlt    surq_gen
        latq    wateryld        perc          et     ecanopy      eplant       esoil   surq_cont
          cn     sw_init    sw_final      sw_ave      sw_300    sno_init   sno_final     snopack
         pet       qtile         irr  surq_runon  latq_runon    overbank    surq_cha    surq_res
     surq_ls    latq_cha    latq_res     latq_ls     gwsoilq       satex  satex_chan   sw_change
     lagsurf     laglatq    lagsatex    wet_evap    wet_oflo    wet_stor   plant_cov       mgt_ops
```

To check an average annual HRU row, compare `precip` + `irr` + `surq_runon` + `latq_runon` with `et` + `wateryld` + `perc` plus the change in soil water and snow storage. A large gap usually points to wetland storage, gwflow exchange or water held in the runoff lag.

The reference outputs in `refdata/Ames_sub1` of the SWAT+ repository were written by revision 61.0.1, and their header has no `plant_cov` and `mgt_ops` names.

Related files

- `print.prt`: switches each file on or off and sets the print window.
- Output naming convention: every output file and the `print.prt` row that controls it.
- Channel and sediment outputs: where the water leaving the landscape goes next.
- Post-processing helpers: reading these files with Python.

Sources

- File opening and header writes: `output_landscape_init.f90`
- Record, header and units types, and the add and average operators: `output_waterbal`, `output_waterbal_header` and `output_waterbal_header_units` in `output_landscape_module.f90`
- Daily values: `hru_control.f90`, with the runoff lag in `sq_surfst.f90` and wetland outflow in `wetland_control.f90`
- Writers: `hru_output.f90`, `hru_lte_output.f90`, `lsu_output.f90`, `basin_output.f90`
- `print.prt` reader: `basin_print_codes_read.f90`
- Print window and step flags: `time_control.f90`

Column	Fortran variable
surq_gen	qday (after sq_surfst)
wateryld	qdr(j)
perc	sepbtm(j)
et	etday
ecanopy	canev
eplant	ep_day
esoil	es_day
surq_cont	surfq(j)
cn	cnday(j)
sw_ave	output_waterbal%sw
pet	pet_day
irr	irrig(j)%applied
surq_runon	ls_overq
latq_runon	latqrunon
overbank	hru(j)%wet_obank_in
gwsoilq	output_waterbal%gwsoil, from gwsoilq(j)
sw_change	output_waterbal%delsw
lagsurf, laglatq, lagsatex	surf_bs(1,j), bss(1,j), bss_ex(1,j)
wet_oflo	output_waterbal%wet_out
wet_stor	wet(j)%flo / (10 * area_ha)

Purpose

These files report what happens in the channel network: flow, sediment and nutrients entering, leaving and stored in each channel, channel geometry and erosion, and sediment budgets for each channel, each stream order and the whole basin.

SWAT+ routes channels defined in the channel files of the SWAT+ Editor (see Channels and channel-lte). Two rows of print.prt control all of the channel files:

print.prt row	Files opened
channel_sd	channel_sd_*, channel_sdmorph_*, sd_chanbud_*, and the sediment budget files ch_sedbud_*, chanbud_order_* and sedbud_*. With a sub-daily time step and daily set to y, also channel_sd_subday.txt.
basin_sd_cha	basin_sd_cha_*, basin_sd_chamorph_*, basin_sd_chanbud_*

Each y in the daily, monthly, yearly or avann column of a row opens the matching file. Most files use the suffixes _day, _mon, _yr and _aa. The ch_sedbud and chanbud_order files use _d, _m, _y and _a. With csvout set to y, a .csv copy of each file is written as well.

The channel files are opened only when the project has at least one channel, except sd_chanbud_*, which is opened whenever its switch is on.

The channel and basin_cha rows open the legacy channel files described at the end of this page.

Files at a glance

File	One row per	Content
channel_sd_*	channel	Water surface, storage, inflow and outflow of water, sediment and nutrients, and stream temperature.
channel_sdmorph_*	channel	Flow rates, sediment transport and erosion, and channel width, depth and slope.

File	One row per	Content
`sd_chanbud_*`	channel	Sediment and nutrient budget: inflow, outflow, deposition and erosion.
`ch_sedbud_*`	channel	Bank and bed erosion, floodplain deposition, channel size.
`chanbud_order_*`	stream order (1 to 12)	The `ch_sedbud` values summed over all channels of each stream order.
`sedbud_*`	basin (one row)	Basin sediment budget: upland yield, bank and bed erosion, floodplain, reservoir and wetland deposition.
`basin_sd_cha_*`	basin	`channel_sd` summed over all channels.
`basin_sd_chamorph_*`	basin	`channel_sdmorph` summed over all channels.
`basin_sd_chanbud_*`	basin	`sd_chanbud` summed over all channels.
`channel_sd_subday.txt`	channel and time step	Outflow for each sub-daily time step.

Leading columns

Files with one row per channel start with seven columns:

#	Column	Description
1	`jday`	Day of the year of the last day in the print step.
2	`mon`	Month.
3	`day`	Day of the month.
4	`yr`	Calendar year.
5	`unit`	Channel number.
6	`gis_id`	GIS identifier of the channel.
7	`name`	Channel name.

The basin files have the same seven columns with `unit` and `gis_id` set to 1 and the project name in `name`. In `ch_sedbud_*` the same three columns are labelled `ichan`, `ob` and `name`. `chanbud_order_*` has `iord` (the stream order) in place of all three, and `sedbud_*` has only the four date columns.

channel_sd files

After the leading columns come four values for the water surface, then three blocks of 18 values (storage, inflow, outflow) and a final stream temperature.

Column	Units	Description
`area`	ha	Water surface area of the channel.
`precip`	m^3	Precipitation falling on the water surface.
`evap`	m^3	Evaporation from the water surface.
`seep`	m^3	Seepage through the channel bed.

Each block has the same 18 quantities. The column names end in `_stor`, `_in` or `_out`:

Quantity	Storage units	Inflow and outflow units	Description
`flo`	m^3	m^3/s	Water: volume stored, or average flow rate.
`sed`	tons	tons	Sediment.
`orgn`	kg N	kg N	Organic nitrogen.
`sedp`	kg P	kg P	Organic (sediment-attached) phosphorus.
`no3`	kg N	kg N	Nitrate.
`solp`	kg P	kg P	Soluble mineral phosphorus.
`chla`	kg	kg	Chlorophyll a.

Quantity	Storage units	Inflow and outflow units	Description
nh3	kg N	kg N	Ammonium.
no2	kg N	kg N	Nitrite.
cbod	kg	kg	Carbonaceous biochemical oxygen demand.
dox	kg	kg	Dissolved oxygen.
san	tons	tons	Sand.
sil	tons	tons	Silt.
cla	tons	tons	Clay.
sag	tons	tons	Small aggregates.
lag	tons	tons	Large aggregates.
grv	tons	tons	Gravel.
null	°C	°C	Water temperature.

The 18th column of each block is headed `null` and has no unit in the units row. It is the water temperature of the stored water, the inflow and the outflow. The outflow temperature is the stream temperature from the heat balance calculation.

The last column, `water_temp` (°C), is not the heat balance temperature. It is $5 + 0.75\,\bar{T}$, where $\bar{T}$ is the mean air temperature of the day at the weather station of the channel that was routed last. All channels therefore show the same value for a given day. Use the `null` column of the outflow block for the stream temperature of each channel.

Aggregation

- The storage block is the storage at the end of the print step, also in the monthly, yearly and average annual files.
- Inflow and outflow rates (`flo_in`, `flo_out`) are averaged over the days of the step. All other inflow and outflow quantities are totals for the step; the average annual file gives the average yearly total.
- Water temperatures in the inflow and outflow blocks are flow-weighted averages.
- `precip`, `evap` and `seep` are totals for the step. `area` is the average daily area in the monthly and yearly files.

channel_sdmorph files

Column	Units	Description
flo_in	m^3/s	Inflow rate.
geo_bf	m^3/s	Base flow from the geomorphic aquifer method.
flo_out	m^3/s	Outflow rate.
peakr	m^3/s	Peak flow rate.
sed_in	tons	Sediment entering the channel.
sed_out	tons	Sediment leaving the channel.
washld	tons	Wash load (suspended sediment) leaving the channel.
bedld	tons	Sediment deposited in the channel.
dep	tons	Sediment deposited on the floodplain.
deg_btm	tons	Bed erosion.
deg_bank	tons	Bank erosion.
hc_sed	tons	Sediment from gully headcut erosion.
width	m	Bankfull top width at the end of the step.
depth	m	Bankfull depth at the end of the step.
slope	m/m	Channel slope at the end of the step.
deg_btm	m	Bed downcutting.
deg_bank	m	Bank widening.
hc_len	m	Headcut retreat.
flo_in_mm	mm	Inflow as a depth over the channel's drainage area.
aqu_in_mm	mm	Geomorphic base flow as a depth over the drainage area.
flo_out_mm	mm	Outflow as a depth over the drainage area.
sed_stor	tons	Sediment stored in the channel.
n_tot	kg N	Total nitrogen leaving the channel (organic N, nitrate, ammonium and nitrite).

Column	Units	Description
p_tot	kg P	Total phosphorus leaving the channel (organic and soluble P).
dep_bf	m	Flow depth at bankfull.
velav_bf	m/s	Average velocity at bankfull.

deg_btm and deg_bank each appear twice, first in tons and then in m; read them by position. The units row labels n_tot as kg_N, p_tot as kg_P and dep_bf as m/s; dep_bf is a depth in m.

In the monthly and yearly files, the flow rates, sed_stor, n_tot, p_tot, dep_bf and velav_bf are daily averages; the tons, mm and m erosion columns are totals; and width, depth and slope are the values at the end of the step. The average annual file gives daily averages for the first group and average yearly totals for the second.

sd_chanbud files

Column	Description
in_sed, out_sed	Sediment entering and leaving the channel.
fp_dep	Sediment deposited on the floodplain.
ch_dep	Sediment deposited in the channel.
bank_ero	Bank erosion.
bed_ero	Bed erosion.
in_no3, in_orgn, out_no3, out_orgn	Nitrate and organic nitrogen entering and leaving.
fp_no3	Nitrate lost on the floodplain.
bank_no3, bed_no3	Nitrate gained from bank and bed erosion.
fp_orgn, ch_orgn	Organic nitrogen deposited on the floodplain and in the channel.
bank_orgn, bed_orgn	Organic nitrogen gained from bank and bed erosion.
in_solp, in_orgp, out_solp, out_orgp	Soluble and organic phosphorus entering and leaving.
fp_solp	Soluble phosphorus lost on the floodplain.
bank_solp, bed_solp	Soluble phosphorus gained from bank and bed erosion.
fp_orgp, ch_orgp	Organic phosphorus deposited on the floodplain and in the channel.
bank_orgp, bed_orgp	Organic phosphorus gained from bank and bed erosion.
no3_orgn	Nitrate converted to organic nitrogen in the channel.
solp_orgp	Soluble phosphorus converted to organic phosphorus in the channel.

The sediment columns (in_sed to bed_ero) are in tons. The nitrogen and phosphorus columns are in kg, although the units row shows tons for every column. All values are totals for the print step; the average annual file gives the average yearly total.

Sediment budget files

These three file groups are new in 62.0.1. They use the same channel erosion and deposition values, summed in different ways. Their rows are written in free format (list-directed), so column widths vary.

ch_sedbud files (per channel)

Column	Units	Description
num	-	Number of days in the step on which the channel had inflow.
wid	m	Bankfull width, summed over those days.
dep	m	Bankfull depth, summed over those days.
fp_km2	km^2	Floodplain area, summed over those days. The floodplain is taken as five times the channel width over the channel length.
ebank_t	tons	Bank erosion.
ebtm_t	tons	Bed erosion.

Column	Units	Description
ebank_m	m	Bank erosion as a width.
ebtm_m	m	Bed erosion as a depth.
fp_t	tons	Floodplain deposition.
fp_mm	mm	Floodplain deposition as a depth over the floodplain, assuming a bulk density of 1 t/m³.

wid, dep and fp_km2 are summed over days, so divide them by num to get the average width, depth and floodplain area in the daily, monthly and yearly files. In the average annual file the sums are divided by the number of printed years but num is not, so the ratio is too small by that factor.

chanbud_order files (per stream order)

One row for each stream order from 1 to 12 in every step, with the ch_sedbud columns summed over all channels of that order. num is then the number of channel-days with inflow, and wid / num and dep / num give the average width and depth for the order. fp_mm is summed over channels and is not a depth for the order as a whole. The stream order of each channel comes from the order column of the channel database.

sedbud files (basin)

Column	Units	Description
upland_t	tons	Sediment yield from all HRUs.
ebank_t	tons	Bank erosion in all channels.
ebtm_t	tons	Bed erosion in all channels.
fp_t	tons	Floodplain deposition in all channels.
res_dep_t	tons	Net deposition in all reservoirs (sediment in minus sediment out).
wet_dep_t	tons	Net deposition in all wetlands.

Deposition inside the channels themselves is not part of this file; it is in the ch_dep column of sd_chanbud.

Print timing

The three sediment budget file groups start after the nyskip warm-up years, like the other files. Their daily files have a row for every day after that and do not follow the daily print window (day_start to day_end, interval) in print.prt.

Basin files

basin_sd_cha_*, basin_sd_chamorph_* and basin_sd_chanbud_* have the same columns as channel_sd_*, channel_sdmorph_* and sd_chanbud_*, with one row for the basin. Each value is the sum over all channels in the basin. Flows are summed along the whole network, so flo_out in basin_sd_cha is not the flow leaving the basin; use the outlet channel's row in channel_sd for that. basin_sd_cha has no water_temp column, and in basin_sd_chamorph the width, depth and slope columns are those of the last channel.

Sub-daily outflow

When the simulation uses a sub-daily time step and the channel_sd row has y in daily, channel_sd_subday.txt has one row per channel per time step: jday, mon, day, yr, unit, gis_id, tstep (time step number), name and flo_out. flo_out is the volume of water (m³) leaving the channel in that time step, although the header gives m³/s. The units row and the header of the .csv copy are those of the channel_sdmorph file.

Legacy channel files

The `channel` and `basin_cha` rows of print.prt open `channel_*.txt` and `basin_cha_*.txt` for legacy channel objects (`chan` in object.cnt). Their 59 columns cover flow, sediment, nutrients, pesticides, bacteria, metals, sediment size classes and erosion:

```
flo_in flo_out evap tloss sed_in sed_out sed_conc orgn_in orgn_out orgp_in orgp_out
no3_in no3_out nh4_in nh4_out no2_in no2_out solp_in solp_out chla_in chla_out
cbod_in cbod_out dis_in dis_out solpst_in solpst_out sorbpst_in sorbpst_out react
volat setlpst resuspst difus reactb bury sedpest bacp baclp met1 met2 met3 sand_in
sand_out silt_in silt_out clay_in clay_out smag_in smag_out lag_in lag_out grvl_in
grvl_out bnk_ero ch_deg ch_dep fp_dep tot_ssed
```

SWAT+ 62.0.1 does not route legacy channels (the routing call is disabled), so these files contain only zeros. Projects built with the SWAT+ Editor use the channels described above.

Example

The columns of every `chanbud_order_*` file:

```
jday  mon  day  yr  iord  num  wid  dep  fp_km2  ebank_t  ebtm_t  ebank_m  ebtm_m  fp_t  fp_mm
```

Take 40 first-order channels, each 5 m wide, that have flow on every day of a 10-year print period. In `chanbud_order_y.txt` the row with `iord` 1 has `num` = 40 × 365 = 14,600 and `wid` = 73,000, so `wid` / `num` gives the 5 m average width. In `chanbud_order_a.txt` the same row has `num` = 146,000 (the count over all ten years) and `wid` = 73,000 (the ten-year sum divided by 10), so the average width is 10 × `wid` / `num`.

The reference outputs in `refdata/Ames_sub1` of the SWAT+ repository were written by revision 61.0.1 and contain none of the sediment budget files.

Related files

- print.prt: switches each file on or off.
- Channels and channel-lte: channel inputs, including the stream `order` column.
- Water balance outputs: the landscape water that reaches the channels.
- Aquifer, reservoir, plant outputs: reservoir and wetland files.
- Output naming convention: all output files.

Sources

- File opening and headers: header_sd_channel.f90, header_write.f90, header_channel.f90
- Header and record types: sd_channel_module.f90, hydrograph_module.f90, water_body_module.f90, channel_module.f90, `basin_sediment_budget` in basin_module.f90
- Channel writers: sd_channel_output.f90, sd_chanmorph_output.f90, sd_chanbud_output.f90
- Sediment budget writers: ch_sedbud_output.f90, chord_sedbud_output.f90, basin_sedbud_output.f90
- Basin writers: basin_sdchannel_output.f90, basin_chanmorph_output.f90, basin_chanbud_output.f90, basin_channel_output.f90
- Values: sd_channel_control3.f90, sd_channel_sediment3.f90, ch_temp.f90; upland, reservoir and wetland terms in hru_control.f90, res_control.f90, wetland_control.f90
- Call order: command.f90

File	Header type	Record
`channel_sd_*`	`ch_wbod_hdr`, `hyd_stor_hdr`, `hyd_in_hdr`, `hyd_out_hdr`, `wtmp_hdr`	`ch_wat_*`, `ch_stor`, `ch_in_*`, `ch_out_*`, `wtemp`
`channel_sdmorph_*`	`sdch_hdr`	`chsd_*` (`sd_ch_output`)
`sd_chanbud_*`	`sdch_bud_hdr`	`ch_sed_bud*` (`channel_sediment_budget_output`)
`ch_sedbud_*`	`ch_sedbud_hdr`	`ch_morph*` (`channel_morphology_output`)
`chanbud_order_*`	`ch_bud_order_hdr`	`ch_morph_ord*`

File	Header type	Record
sedbud_*	ch_sed_bud_hdr	bsn_sedbud* (basin_sediment_budget)
channel_sd_subday.txt	sdch_hdr_subday	ob%hyd_flo(1,ii)

Column	Fortran variable
geo_bf	sd_ch_output%aqu_in
hc_len	sd_ch_output%hc_m
flo_out_mm	sd_ch_output%flo_mm
second deg_btm, deg_bank	deg_btm_m, deg_bank_m
null (each block)	hyd_output%temp
water_temp	module variable wtemp
ob (ch_sedbud)	ob(iob)%gis_id

Purpose

This page covers four groups of output files:

- aquifer storage and fluxes (aquifer_*.txt, basin_aqu_*.txt)
- reservoir storage and fluxes (reservoir_*.txt, basin_res_*.txt)
- crop yields (crop_yld_*.txt, basin_crop_yld_*.txt)
- plant growth, stress and weather (hru_pw_*.txt, hru-lte_pw_*.txt, lsunit_pw_*.txt, basin_pw_*.txt)

Each file is switched on in print.prt. The suffix gives the print interval: _day, _mon, _yr or _aa (average annual). When csvout is y, a .csv copy with the same columns is written beside each .txt file. The first seven columns of every file except the crop yield files are jday, mon, day, yr, unit, gis_id and name; see Naming convention.

Aquifer

Files and switches

File	print.prt row	Scope
aquifer_day.txt, aquifer_mon.txt, aquifer_yr.txt, aquifer_aa.txt	aquifer	one row per aquifer per interval
basin_aqu_day.txt, basin_aqu_mon.txt, basin_aqu_yr.txt, basin_aqu_aa.txt	basin_aqu	one row per interval for the whole basin

The aquifer files report the one-dimensional aquifers defined in aquifer.aqu and are opened only when the model has such aquifers. When the gwflow module replaces the aquifers, use the gwflow outputs instead.

Columns

Line 1 is the project name and version, line 2 the column header and line 3 the units. After the seven leading columns come 17 value columns:

#	Column	Units	Description
1	flo	mm	return flow leaving the aquifer
2	dep_wt	m	depth from the ground surface to the water table
3	stor	mm	water stored in the aquifer
4	rchrg	mm	recharge entering the aquifer from other objects

#	Column	Units	Description
5	`seep`	mm	deep seepage from the bottom of the aquifer
6	`revap`	mm	water removed by plant uptake and evaporation (revap)
7	`no3_st`	kg N/ha	nitrate-N stored in the aquifer
8	`minp`	kg P	mineral P sent out with the return flow (see the notes)
9	`orgn` (header)	%	organic carbon content of the aquifer (see the notes)
10	`orgp` (header)	kg N/ha	organic N in the aquifer (see the notes)
11	`no3_rchg`	kg N/ha	nitrate-N entering with recharge; `rchrgn` in earlier documentation
12	`no3_loss`	kg N/ha	nitrate-N lost in the aquifer (decay); `nloss` in earlier documentation
13	`no3_lat`	kg N/ha	nitrate-N leaving with the return flow; `no3gw` in earlier documentation
14	`no3_seep`	kg N/ha	nitrate-N leaving with deep seepage; `seepno3` in earlier documentation
15	`flo_cha`	mm	part of `flo` routed to channels
16	`flo_res`	mm	part of `flo` routed to reservoirs
17	`flo_ls`	mm	part of `flo` routed to other aquifers

Notes on individual columns:

- Columns 9 and 10 are mislabelled. The header says `orgn` (kg/ha_N) and `orgp` (kg/ha_P), but column 9 is the organic carbon content from the `carbon` column of `aquifer.aqu`, constant through the run, and column 10 is organic N, which 62.0.1 never updates, so it stays zero.
- `minp` is the mineral P concentration from the `sol_p` column of `aquifer.aqu` multiplied by the day's recharge volume. The value is a mass in kg for the whole aquifer; the units line gives kg/ha_P. It is computed from the recharge volume and leaves the aquifer with the return flow.
- `flo_cha`, `flo_res` and `flo_ls` split `flo` by the outflow fractions in the aquifer's connection file. Only outflows of type `tot` are counted, so the three columns add up to `flo` only when every outflow is of that type.

Aggregation

Fluxes are summed over the interval. Three state columns are averaged instead: the monthly file divides `stor`, `dep_wt` and `no3_st` by the number of days in the month, and the yearly file divides the sum of the monthly averages by 12. The average annual file divides the yearly values by the number of printed years. Column 9 is never summed: every interval reports the starting value.

The basin file is the area-weighted average of all aquifers, each weighted by its share of the basin area.

Example header

`basin_aqu_yr.txt` from the `Ames_sub1` reference dataset run with 62.0.1:

```
   jday   mon   day    yr   unit  gis_id  name                      flo          dep_wt            stor
                                                                     mm               m              mm
```

Reservoir

Files and switches

File	`print.prt` row	Scope
`reservoir_day.txt`, `reservoir_mon.txt`, `reservoir_yr.txt`, `reservoir_aa.txt`	`reservoir`	one row per reservoir per interval

File	print.prt row	Scope
basin_res_day.txt, basin_res_mon.txt, basin_res_yr.txt, basin_res_aa.txt	basin_res	all reservoirs added together

The reservoir files are opened only when the model has reservoirs.

Columns

After the seven leading columns come four blocks: 4 water body columns, then 18 storage, 18 inflow and 18 outflow columns.

Block	Columns	Units
Water body	area, precip, evap, seep	ha, m³, m³, m³
Storage at the end of the interval	flo_stor, sed_stor, orgn_stor, sedp_stor, no3_stor, solp_stor, chla_stor, nh3_stor, no2_stor, cbod_stor, dox_stor, san_stor, sil_stor, cla_stor, sag_stor, lag_stor, grv_stor, null	see below
Inflow during the interval	the same 18 names ending in _in	see below
Outflow during the interval	the same 18 names ending in _out	see below

Units of the 18 constituent columns, which repeat in the storage, inflow and outflow blocks:

Constituent	Units	Description
flo	m³	water volume
sed	t	sediment
orgn	kg N	organic N
sedp	kg P	organic (sediment-attached) P
no3	kg N	nitrate-N
solp	kg P	soluble mineral P
chla	kg	chlorophyll-a
nh3	kg N	ammonium-N
no2	kg N	nitrite-N
cbod	kg	carbonaceous biochemical oxygen demand
dox	kg	dissolved oxygen
san, sil, cla, sag, lag, grv	t	sand, silt, clay, small aggregates, large aggregates and gravel
null	°C	water temperature (the header gives null and no unit)

The inflow columns are flo_in, sed_in, orgn_in, sedp_in, no3_in, solp_in, chla_in, nh3_in, no2_in, cbod_in, dox_in, san_in, sil_in, cla_in, sag_in, lag_in, grv_in and null. The outflow columns are flo_out, sed_out, orgn_out, sedp_out, no3_out, solp_out, chla_out, nh3_out, no2_out, cbod_out, dox_out, san_out, sil_out, cla_out, sag_out, lag_out, grv_out and null (the last column of each block is headed null).

Inflow and outflow are volumes and masses for the interval, in m³, t and kg. The storage block is the reservoir content when the line is written, so in the monthly, yearly and average annual files it is the storage on the last day of the interval.

Aggregation

Inflow, outflow, precipitation, evaporation and seepage are summed over the interval, and the average annual file divides the totals by the number of printed years. The surface area is averaged over the days of the month, and the yearly value is the average of the monthly values.

reservoir_mon.csv has one extra line after the units line, RES reservoir_mon.csv, which the other reservoir files do not have.

Crop yield

Files and switches

The `crop_yld` column of `print.prt` controls all crop yield files: `y` gives yearly files, `a` average annual files, `b` both, and `n` none.

File	Written when `crop_yld` is	Content
`crop_yld_aa.txt`	`a` or `b`	one row per HRU and plant at the end of the run
`crop_yld_yr.txt`	`y` or `b`	header lines only (see below)
`basin_crop_yld_yr.txt`	`y` or `b`	one row per plant per year for the whole basin
`basin_crop_yld_aa.txt`	`y` or `b`	one row per plant at the end of the run

With `crop_yld = a`, only `crop_yld_aa.txt` is produced; the basin files need `y` or `b`. In 62.0.1 no data rows are written to `crop_yld_yr.txt` (or its `.csv`). For yearly yields use `basin_crop_yld_yr.txt`, or `yield.out` (switched on by `mgtout`) for every harvest.

HRU columns

`crop_yld_aa.txt` has a title line, a line `--YIELD (kg/ha)--`, the column header, and one row per plant in each HRU:

#	Column	Units	Description
1	`jday`	-	day of year of the last simulated day
2	`mon`	-	month
3	`day`	-	day of month
4	`yr`	-	year
5	`unit`	-	HRU number
6	`PLANTNM`	-	plant name from `plants.plt`
7	`MASS`	kg/ha	dry mass of the harvested yield
8	`C`	kg/ha	carbon in the yield
9	`N`	kg/ha	nitrogen in the yield
10	`P`	kg/ha	phosphorus in the yield

The values are the total yield over the whole run divided by the number of harvests of that plant, that is, the average yield per harvest. A plant that was never harvested reports zero. The `.csv` header lists a single `yield` column, while each data row contains the four values `MASS`, `C`, `N` and `P`.

Basin columns

#	Column	Units	Description
1	`year`	-	calendar year (the last year in the `_aa` file)
2	`plant_no`	-	plant number in the basin plant list
3	`plant_name`	-	plant name
4	`harv_area(ha)`	ha	area harvested; an HRU harvested twice in a year counts twice
5	`yld(t)`	t	total dry yield
6	`yld(t/ha)`	t/ha	`yld(t)` divided by `harv_area(ha)`

The yearly file skips the years before `nyskip`. In the average annual file, area and yield are divided by the number of printed years, and `yld(t/ha)` is total yield divided by total harvested area.

Example

`crop_yld_aa.txt` and `basin_crop_yld_aa.txt` from a two-year run of the `Ames_sub1` reference dataset with 62.0.1:

```
                                                                    --YIELD (kg/ha)--
  jday    mon    day     yr     unit PLANTNM                    MASS          C           N           P
   366     12     31   1976        1    corn                                3692.158    1661.471
   366     12     31   1976        1    soyb                                   0.000       0.000

      year       plant_no   plant_name       harv_area(ha)     yld(t)          yld(t/ha)
       1976              1 corn                                  12.04350       47.48362       3.942675
       1976              2 soyb                                0.0000000E+00  0.0000000E+00  0.0000000
```

Plant and weather

Files and switches

File	print.prt row	Scope
`hru_pw_day.txt` to `hru_pw_aa.txt`	`hru_pw`	one row per HRU
`hru-lte_pw_day.txt` to `hru-lte_pw_aa.txt`	`hru-lte_pw`	one row per HRU-lte
`lsunit_pw_day.txt` to `lsunit_pw_aa.txt`	`lsunit_pw`	one row per landscape unit
`basin_pw_day.txt` to `basin_pw_aa.txt`	`basin_pw`	whole basin

All use the columns below. The HRU files add `plant_cov` and `mgt_ops` at the end of each row: the plant community and the management schedule of the HRU's land use.

Columns

#	Column	Units	Description
1	`lai`	m^2/m^2	leaf area index
2	`bioms`	kg/ha	total plant biomass
3	`yield`	kg/ha	harvested yield (dry mass)
4	`residue`	kg/ha	surface residue
5	`sol_tmp`	°C	temperature of soil layer 2
6	`strsw`	days	water (drought) stress days
7	`strsa`	days	aeration (excess water) stress days
8	`strstmp`	days	temperature stress days
9	`strsn`	days	nitrogen stress days
10	`strsp`	days	phosphorus stress days
11	`strss`	days	salinity stress days
12	`nplt`	kg N/ha	nitrogen in the plant biomass
13	`percn`	kg N/ha	nitrate-N leached below the soil profile
14	`pplnt`	kg P/ha	phosphorus in the plant biomass
15	`tmx`	°C	maximum air temperature
16	`tmn`	°C	minimum air temperature
17	`tmpav`	°C	mean air temperature
18	`solarad`	MJ/m^2	solar radiation
19	`wndspd`	m/s	wind speed
20	`rhum`	fraction	relative humidity
21	`phubas0`	°C	accumulated base-zero heat units
22	`lai_max`	m^2/m^2	maximum leaf area index (always 0 in 62.0.1)
23	`bm_max`	kg/ha	maximum total biomass during the interval
24	`bm_grow`	kg/ha	biomass growth
25	`c_gro`	kg C/ha	plant carbon growth

The units line writes `kgha` for columns 2 to 4 and 12 to 14 and `----` for the stress columns.

Aggregation

Columns	Monthly	Yearly	Average annual
`yield`, stress days, `percn`	sum	sum	yearly sum averaged over the printed years
`nplt`, `pplnt`	plant N and P at the end of the month	at the end of the year	at the end of the run
`bm_max`	maximum in the month	maximum in the year	maximum in the run
all other columns	daily average	daily average	daily average

`nplt` and `pplnt` are the N and P contents of the standing plants. They do not report uptake during the interval. After a harvest or kill they drop to zero, so yearly and average annual values are often zero for annual crops.

Example

`hru_pw_mon.txt`, `Ames_sub1` run with 62.0.1 (header, units and first data row):

jday	mon	day	yr	unit	gis_id	name	lai	bioms	yield	residue
							m**2/m**2	kgha	kgha	kgha
31	1	31	1975	1	1	hru0001	0.000	0.000	0.000	17278.244

Related files

- print.prt: switches for every file on this page.
- aquifer.aqu and reservoir.res: the objects reported here.
- Water balance outputs and Channel and sediment outputs.
- gwflow outputs: groundwater output when gwflow replaces the aquifers.

Sources

- Aquifer: aquifer_output.f90 (units 2520 to 2527), basin_aquifer_output.f90 (units 2090 to 2097), headers and arithmetic in aquifer_module.f90, file opening in header_aquifer.f90 and header_write.f90, values set in aqu_1d_control.f90 and aqu_initial.f90
- Reservoir: reservoir_output.f90 (units 2540 to 2547), basin_reservoir_output.f90, header_reservoir.f90, water body type in water_body_module.f90, constituent headers in hydrograph_module.f90
- Crop yield: hru_output.f90 (unit 4008), file opening in output_landscape_init.f90 (units 4008 to 4011) and header_yield.f90, basin rows in time_control.f90 (units 5100 and 5101)
- Plant and weather: hru_output.f90 (units 2040 to 2047), daily values in hru_control.f90, type, header and operators in output_landscape_module.f90
- `print.prt` rows: basin_print_codes_read.f90

Output column	Fortran variable
aquifer column 9 (`orgn` in the header)	`aqu_d%cbn` (from `aqu_dat%cbn`)
aquifer column 10 (`orgp` in the header)	`aqu_d%orgn`
aquifer `minp`	`aqu_d%minp`
reservoir water body block	`res_wat_d` (`type water_body`)
reservoir storage block	`res(j)` (`type hyd_output`)
reservoir inflow and outflow blocks	`res_in_d`, `res_out_d`
`crop_yld_aa` `MASS`, `C`, `N`, `P`	`pl_mass%yield_tot(ipl)` divided by `pcom%plcur(ipl)%harv_num`
plant and weather `nplt`, `pplnt`	`hpw_d%nplnt`, `hpw_d%pplnt`, both copied from `pl_mass%tot_com`
plant and weather `solarad`, `phubas0`	`hpw_d%solrad`, `hpw_d%phubase0`

Purpose

Two groups of output files report nutrients on the land:

- The nutrient balance files (`*_nb_*`) give the N and P added to, moved within and removed from the soil and plants: grazing and fertiliser inputs, fixation, atmospheric deposition, mineralisation and other transfers between soil pools, denitrification and plant uptake.
- The losses files (`*_ls_*`) give the sediment and nutrients that leave the land with surface runoff, lateral flow, tile flow and percolation.

All values are amounts per unit area summed over the print interval. Each group has a row in `print.prt` for each scope, with daily, monthly, yearly and average annual flags.

Files and switches

Scope	Nutrient balance row	Nutrient balance files	Losses row	Losses files
HRU	`hru_nb`	`hru_nb_{day,mon,yr,aa}.txt`	`hru_ls`	`hru_ls_{day,mon,yr,aa}.txt`
HRU-lte	`hru-lte_nb`	none (see below)	`hru-lte_ls`	`hru-lte_ls_{day,mon,yr,aa}.txt`
Landscape unit	`lsunit_nb`	`lsunit_nb_{day,mon,yr,aa}.txt`	`lsunit_ls`	`lsunit_ls_{day,mon,yr,aa}.txt`
Basin	`basin_nb`	`basin_nb_{day,mon,yr,aa}.txt`	`basin_ls`	`basin_ls_{day,mon,yr,aa}.txt`

Each file also gets a `.csv` copy when `csvout = y`. Notes on the switches:

- `print.prt` also accepts `hru-lte_nb`, `region_nb` and `region_ls` rows, but 62.0.1 writes no files for them.
- The `hru_nb` row also opens `hru_ncycle_{day,mon,yr,aa}.txt`, a nutrient cycling file with the columns `lab_min_p`, `act_sta_p`, `act_nit_n`, `act_sta_n`, `org_lab_p`, `rsd_hs_c`, `rsd_nitrorg_n` and `rsd_laborg_p`. In 62.0.1 these files contain only their header lines. The same transfers are in the `hru_nb` files.
- Landscape unit files are written only when the model defines landscape units in `ls_unit.def`.

Line 1 of every file is the project name and version, line 2 the column header and line 3 the units. The first seven columns are `jday`, `mon`, `day`, `yr`, `unit`, `gis_id` and `name`; see Naming convention.

Nutrient balance columns

#	Column	Units	Description
1	`grzn`	kg N/ha	nitrogen added to the soil in manure from grazing
2	`grzp`	kg P/ha	phosphorus added to the soil in manure from grazing
3	`lab_min_p`	kg P/ha	net phosphorus flow from the labile mineral pool to the active mineral pool; negative when the flow goes the other way
4	`act_sta_p`	kg P/ha	net phosphorus flow from the active mineral pool to the stable mineral pool; negative when the flow goes the other way
5	`fertn`	kg N/ha	nitrogen applied with fertiliser and manure (mineral and organic)
6	`fertp`	kg P/ha	phosphorus applied with fertiliser and manure (mineral and organic)
7	`fixn`	kg N/ha	nitrogen added to plant biomass by fixation
8	`denit`	kg N/ha	nitrogen lost from the nitrate pool by denitrification
9	`act_nit_n`	kg N/ha	nitrogen moved from the active organic pool to the nitrate pool (mineralisation)
10	`act_sta_n`	kg N/ha	nitrogen moved from the active organic pool to the stable organic pool
11	`org_lab_p`	kg P/ha	phosphorus moved from the organic pool to the labile pool (mineralisation)
12	`rsd_nitorg_n`	kg N/ha	nitrogen moved from fresh residue to the nitrate (80%) and active organic (20%) pools

#	Column	Units	Description
13	rsd_laborg_p	kg P/ha	phosphorus moved from fresh residue to the labile (80%) and organic (20%) pools
14	no3atmo	kg N/ha	nitrate added to the soil by atmospheric deposition, wet and dry
15	nh4atmo	kg N/ha	ammonium added to the soil by atmospheric deposition, wet and dry
16	nuptake	kg N/ha	nitrogen taken up by plants
17	puptake	kg P/ha	phosphorus taken up by plants
18	gwsoiln	kg N/ha	nitrate added to the soil from groundwater (gwflow module only)
19	gwsoilp	kg P/ha	phosphorus added to the soil from groundwater (gwflow module only)

The units line writes kgha for every column. The HRU files add plant_cov and mgt_ops at the end of each row: the plant community and management schedule of the HRU's land use. The landscape unit and basin files have the same two labels in their header but write no values under them.

Columns 9 to 13 (act_nit_n to rsd_laborg_p) are filled only with the static carbon model, carbon = 0 in codes.bsn. With carbon = 2 they stay at zero, and the corresponding flows are in the carbon outputs. denit and the mineral P columns 3 and 4 are filled under both settings.

Losses columns

#	Column	Units	Description
1	sedyld	t/ha	sediment yield leaving the land by water erosion
2	sedorgn	kg N/ha	organic nitrogen attached to sediment in surface runoff
3	sedorgp	kg P/ha	organic phosphorus attached to sediment in surface runoff
4	surqno3	kg N/ha	nitrate in surface runoff
5	lat3no3	kg N/ha	nitrate in lateral flow
6	surqsolp	kg P/ha	soluble phosphorus in surface runoff
7	usle	t/ha	soil loss predicted by the USLE (the units line says tons)
8	sedminp	kg P/ha	mineral phosphorus (active and stable) attached to sediment
9	tileno3	kg N/ha	nitrate in tile flow
10	lchlabp	kg P/ha	soluble (labile) phosphorus leached below the soil profile
11	tilelabp	kg P/ha	soluble (labile) phosphorus in tile flow
12	satexn	kg N/ha	nitrate in saturation excess runoff; always 0 in 62.0.1

The HRU files add plant_cov, mgt_ops and percn at the end of each row. percn (kg N/ha) is the nitrate leached below the soil profile, the same value as percn in the plant and weather files. The HRU-lte, landscape unit and basin files have these three labels in their header but write no values under them.

In the HRU-lte files only sedyld is calculated; the other columns are always zero.

Aggregation

Every column is a sum. The monthly and yearly files add up the daily values, and the average annual file divides the run total by the number of printed years. Nothing is averaged over days.

Landscape unit values are the sum of the HRU (and HRU-lte) values, each multiplied by its RU_FRAC in ls_unit.ele. Basin values are the sum of the HRU and HRU-lte values, each multiplied by its BSN_FRAC in ls_unit.ele. The fractions are used as given and are not checked: when they do not add up to 1, the basin and landscape unit values

are scaled by the same factor. In the `Ames_sub1` reference dataset the `BSN_FRAC` values add up to 2, so its basin values are twice the area-weighted HRU average.

Example

`hru_nb_yr.txt` and `hru_ls_yr.txt` from a two-year run of the `Ames_sub1` reference dataset with 62.0.1, header, units and first data row:

jday	mon	day	yr	unit	gis_id	name	grzn	grzp	lab_min_p	act_sta_p
							kgha	kgha	kgha	kgha
365	12	31	1975	1	1	hru0001	0.000	0.000	911.096	721.581

jday	mon	day	yr	unit	gis_id	name	sedyld	sedorgn	sedor
							tha	kgha	kg
365	12	31	1975	1	1	hru0001	0.033	0.172	0.

This dataset runs the dynamic carbon model (`carbon = 2`), so columns 9 to 13 of the nutrient balance are zero.

Related files

- print.prt: output switches.
- Water balance outputs: the runoff, lateral flow, tile flow and percolation volumes.
- Aquifer, reservoir, and plant outputs: plant N and P and leached nitrate.
- Carbon outputs: soil carbon and the N and P in the carbon pools.
- Nutrient cycling theory.

Sources

- Column types and headers: `type output_nutbal`, `type output_losses`, `output_nutbal_header`, `output_losses_header` and their units types in output_landscape_module.f90
- File opening: output_landscape_init.f90 (HRU units 2020 to 2037, `hru_ncycle` 3333 to 3340, basin 2060 to 2077, landscape unit 2150 to 2167, HRU-lte losses 2440 to 2447)
- Writers: hru_output.f90, hru_lte_output.f90, lsu_output.f90, basin_output.f90
- Daily values: hru_control.f90, hru_lte_control.f90, nut_pminrl.f90, nut_nminrl.f90, nut_nrain.f90, nut_solp.f90, pl_nup.f90, pl_pup.f90, ero_ysed.f90, pl_manure.f90, rsd_decomp.f90, cbn_zhang2.f90
- Weights: `BSN_FRAC` and `RU_FRAC` read in lsu_read_elements.f90
- `print.prt` rows: basin_print_codes_read.f90

Output column	Fortran variable
`grzn`, `grzp`	`hnb_d%grazn`, `hnb_d%grazp`
`lat3no3`	`hls_d%latno3`
`sedminp`	`hls_d%sedminp = sedminpa + sedminps`
`sedyld`	`sedyld / hru%area_ha`
`percn` (losses, HRU)	`hpw_d%percn` (`percn_aa = hpw_a%percn / yrs_prt` in the `_aa` file)
landscape unit rows	`runb_*`, `ruls_*` built with `lsu_elem%ru_frac`
basin rows	`bnb_*`, `bls_*` built with `lsu_elem%bsn_frac`

Purpose

SWAT+ moves water and its loads between objects (HRUs, routing units, channels, reservoirs, aquifers and so on) as hydrographs: a record of water volume, sediment, nutrients and temperature for the day. The files on this page report those hydrographs.

File	One row per	Content
`hydout_*`	object and outgoing connection	What each object sends to each receiving object.
`hydin_*`	object and incoming connection	What each object receives from each source object.
`deposition_*`	HRU or routing unit that receives inflow	Material deposited from incoming water.
`ru_*`	routing unit	Sum of the hydrographs of the routing unit's elements.
`hydcon.out`	object	The routing order and connections, written once at the start of the run.

`hydin`, `hydout` and `deposition` are switched on by the `hyd` row of `print.prt`, and `ru` by the `ru` row. Each `y` in the `daily`, `monthly`, `yearly` or `avann` column opens the `_day`, `_mon`, `_yr` or `_aa` file. `hydcon.out` is switched on by `hydcon` on the line after `csvout`. With `csvout` set to `y`, a `.csv` copy of each file is written as well.

The connections themselves are defined in the connection files (see connection files): each object lists the objects it sends water to, the hydrograph type it sends and the fraction.

Hydrograph columns

All four time-step file groups end with the same 18 columns:

Column	Units	Description
`flo`	m^3	Volume of water.
`sed`	tons	Sediment.
`orgn`	kg N	Organic nitrogen.
`sedp`	kg P	Organic (sediment-attached) phosphorus.
`no3`	kg N	Nitrate.
`solp`	kg P	Soluble mineral phosphorus.
`chla`	kg	Chlorophyll a.
`nh3`	kg N	Ammonium.
`no2`	kg N	Nitrite.
`cbod`	kg	Carbonaceous biochemical oxygen demand.
`dox`	kg	Dissolved oxygen.
`san`	tons	Sand.
`sil`	tons	Silt.
`cla`	tons	Clay.
`sag`	tons	Small aggregates.
`lag`	tons	Large aggregates.
`grv`	tons	Gravel.
`null`	°C	Water temperature.

`flo` is a volume: m^3 per day in the daily file, the total for the month or year in the monthly and yearly files, and the average yearly total in the average annual file. The units row of the files gives m^3/s for `flo`. To convert a daily value to an average rate, divide by 86,400.

All other loads are totals for the step in the same way. The last column is headed `null` and is the water temperature; monthly, yearly and average annual values are flow-weighted averages.

hydout files

#	Column	Description
1	`jday`	Day of the year of the last day in the print step.
2	`mon`	Month.
3	`day`	Day of the month.

#	Column	Description
4	`yr`	Calendar year.
5	`name`	Name of the source object.
6	`type`	Object type of the source object (`hru`, `hru_lte`, `ru`, `aqu`, `chandeg`, `res`, `recall`, `exco`, `dr`, `outlet`).
7	`objtyp`	Type code of the receiving object, as in the connection file (`hru`, `ru`, `aqu`, `sdc`, `res` and so on).
8	`typ_no`	Number of the receiving object within its type.
9	`hyd_typ`	Hydrograph type sent: `tot` (total), `rhg` (recharge), `sur` (surface runoff), `lat` (lateral flow) or `til` (tile flow).
10	`fraction`	Fraction of that hydrograph sent to this receiving object.
11 to 28	`flo` to `null`	The hydrograph sent, already multiplied by `fraction`.

An object with three outgoing connections has three rows per print step. The rows of all objects follow the routing order, so the rows of one step are not sorted by object name.

hydin files

The columns are those of `hydout`, seen from the receiving side:

#	Column	Description
5	`name`	Name of the receiving object.
6	`type`	Object type of the receiving object.
7	`objtyp`	Type code of the source object.
8	`typ_no`	Number of the source object within its type.
9	`hyd_typ`	Hydrograph type received.
10	`fraction`	Fraction of the source hydrograph received.
11 to 28	`flo` to `null`	The hydrograph received.

Rows are written for every incoming connection, but values are filled only when the receiving object is not an HRU, HRU-LTE or routing unit. For those three receiving types the incoming water is split into surface, lateral and tile parts and routed inside the object, and the `hydin` rows stay zero. Use the `hydout` rows of the source object instead.

`hydin_yr.csv` has one more column than the other `hydin` files: the object number is written after `type`, so the later columns are shifted by one against the header.

deposition files

`deposition_*` has `jday`, `mon`, `day`, `yr`, `name` and `type`, then the 18 hydrograph columns. It has a row for every HRU and routing unit that receives water from another object.

- For an HRU without a wetland, the `sed` column is the sediment deposited on the HRU from incoming surface runoff, in tons: the incoming sediment above the transport capacity of the runoff over the HRU. The other columns are zero.
- For an HRU with a wetland or rice paddy, incoming surface runoff and tile flow go into the wetland, and the row shows that inflow. In this case `flo` is in mm and the loads are per hectare.
- For a routing unit, the row shows the hydrograph of the last element processed in the routing unit. It does not measure deposition.

The deposition files are written from the start of the simulation, including the `nyskip` warm-up years, while the average annual value is divided by the number of printed years only.

ru files

`ru_*` has `jday`, `mon`, `day`, `yr`, `name` and `type`, then the 18 hydrograph columns, with one row per routing unit.

Each value is the sum, over the elements of the routing unit, of every hydrograph type the element produces. For an HRU element that is the total hydrograph plus its recharge, surface, lateral and tile hydrographs, so `flo` and the loads are larger than the routing unit's outflow. The outflow of a routing unit is in the `hydout` rows of the routing unit, with `hyd_typ` `tot` for the total flow.

hydcon.out

`hydcon.out` has no title or header line. It lists the objects in the order SWAT+ processes them, one line per object:

Field	Description
1	Sequence number in the routing order.
2	Object name.
3	Object type.
4	Number of the object's properties record in its data file.
5	Overbank connection pointer to landscape units (0 when none).
6	Number of outgoing connections.
7	Number of incoming connections.
8 and on	For each outgoing connection: the receiving object numbers, the receiving type code, its number within the type and the hydrograph type.

In 62.0.1 the receiving object numbers are written as the full list of all receiving objects for every outgoing connection, so lines for objects with more than one outgoing connection repeat that list.

Constituent hydrographs

print.prt and the source refer to hydrograph files for pesticides, pathogens, metals and salts (`hydin_pests_*`, `hydout_salts_*` and so on). SWAT+ 62.0.1 does not write them.

Print timing

`hydin`, `hydout` and `ru` start after the `nyskip` warm-up years. Daily rows follow the daily print window and `interval` in print.prt. `deposition` is written from the first simulated year, as described above.

Example

The column names of `hydout_day.txt` and `hydin_day.txt`:

```
jday  mon  day  yr  name  type  objtyp  typ_no  hyd_typ  fraction
flo  sed  orgn  sedp  no3  solp  chla  nh3  no2  cbod  dox  san  sil  cla  sag  lag  grv  null
```

A `hydout` row for an HRU sending 60% of its total flow to a routing unit has `type` `hru`, `objtyp` `ru`, `hyd_typ` `tot` and `fraction` 0.6, and its `flo` is 0.6 times the HRU's water yield in m^3.

Related files

- print.prt: the `hyd`, `ru` and `hydcon` switches.
- Connection files: the connections these files report.
- Recall outputs: point source and inlet hydrographs, in the same column layout.
- Channel and sediment outputs: channel inflow and outflow as rates.
- Output naming convention: all output files.

Sources

- File opening and headers: header_hyd.f90 (`hydin`, `hydout`, `deposition`, `hydcon.out`), header_write.f90 (`ru`)

- Header types: `hyd_header`, `hyd_header_time`, `hyd_header_obj`, `hyd_header_units`, `hyd_header_units2` in hydrograph_module.f90
- Writers: hydout_output.f90, hydin_output.f90, hyddep_output.f90, ru_output.f90, hyd_connect_out.f90
- Values: command.f90 (incoming and outgoing hydrographs), ru_control.f90, hru_control.f90, rls_routesurf.f90, hru_hyds.f90
- Hydrograph type codes: hyd_connect.f90

Column	Fortran variable
`type`	`ob(icmd)%typ`
`objtyp`, `typ_no`, `hyd_typ`, `fraction` (`hydout`)	`ob(icmd)%obtyp_out(iout)`, `obtypno_out`, `htyp_out`, `frac_out`
`objtyp`, `typ_no`, `hyd_typ`, `fraction` (`hydin`)	`ob(icmd)%obtyp_in(iin)`, `obtypno_in`, `htyp_in`, `frac_in`
hydrograph (`hydout`)	`ht1 = frac_out * ob(icmd)%hd(ihtyp_out)`
hydrograph (`hydin`)	`ob(icmd)%hin_d(iin)` and the `hin_m`, `hin_y`, `hin_a` sums
hydrograph (`deposition`)	`ht1` after `hru_control` or `ru_control`, summed into `ob(icmd)%hdep_m`, `hdep_y`, `hdep_a`
hydrograph (`ru`)	`ru_d(iru)` and the `ru_m`, `ru_y`, `ru_a` sums
`null`	`hyd_output%temp`

Purpose

A recall object adds a measured or externally simulated time series of water and loads to the model: a point source such as a treatment plant, an inlet from an area outside the model, or, with negative flows, a diversion. The recall output files show what each recall object released into the model.

print.prt row	Files	One row per
`recall`	`recall_day.txt`, `recall_mon.txt`, `recall_yr.txt`, `recall_aa.txt`	recall object
`basin_psc`	`basin_psc_day.txt`, `basin_psc_mon.txt`, `basin_psc_yr.txt`, `basin_psc_aa.txt`	basin (sum of all recall objects)

Each `y` in the `daily`, `monthly`, `yearly` or `avann` column of print.prt opens the matching file. With `csvout` set to `y`, a `.csv` copy of each file is written as well. The `basin_psc` files are written only when the project has at least one recall object.

Format

- Line 1: the project name from file.cio and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units.
- Lines 4 and on: data rows.

#	Column	Units	Description
1	`jday`	-	Day of the year of the last day in the print step.
2	`mon`	-	Month.
3	`day`	-	Day of the month.
4	`yr`	-	Calendar year.
5	`name`	-	Name of the recall object. In `basin_psc` files, the project name.

#	Column	Units	Description
6	`type`	-	Object type, always `recall`. `basin_psc` files have no `type` column; the header has a blank name in its place.
7	`flo`	m^3	Water released.
8	`sed`	tons	Sediment.
9	`orgn`	kg N	Organic nitrogen.
10	`sedp`	kg P	Organic (sediment-attached) phosphorus.
11	`no3`	kg N	Nitrate.
12	`solp`	kg P	Soluble mineral phosphorus.
13	`chla`	kg	Chlorophyll a.
14	`nh3`	kg N	Ammonium.
15	`no2`	kg N	Nitrite.
16	`cbod`	kg	Carbonaceous biochemical oxygen demand.
17	`dox`	kg	Dissolved oxygen.
18	`san`	tons	Sand.
19	`sil`	tons	Silt.
20	`cla`	tons	Clay.
21	`sag`	tons	Small aggregates.
22	`lag`	tons	Large aggregates.
23	`grv`	tons	Gravel.
24	`null`	°C	Water temperature.

The values are the recall record used on the day, unchanged: SWAT+ does not convert the numbers read from the recall time series. The model passes `flo` on to the receiving object as a daily volume in m^3, so that is how the column should be read, although the units row gives m^3/s. The `.csv` copies of the `recall` files use a different units row, which gives `flo` as m^3.

Values by recall time step

The time step of each recall object is set in its recall file:

- Daily: each day uses the record for that day. Outside the years covered by the time series, the values are zero.
- Monthly: the record for the month is used on every day of that month. A monthly row of `recall_mon.txt` is therefore the monthly record multiplied by the number of days in the month. Outside the covered years, the values are zero.
- Yearly: the record for the year is used on every day of that year, so a row of `recall_yr.txt` is 365 or 366 times the yearly record.
- Sub-daily: the daily row is the sum of the records for that day's time steps.

A negative `flo` in a daily recall time series is a withdrawal. The row shows the negative value as read. In addition, SWAT+ removes nitrate, ammonium, nitrite, organic nitrogen, soluble phosphorus and dissolved oxygen from the receiving channel's storage at the channel's current concentration. Those removed amounts are not in the recall files.

Aggregation

Monthly and yearly rows are totals over the step for all columns except `null`, which is a flow-weighted average. The average annual row is the average yearly total. Rows start after the `nyskip` warm-up years, and daily rows follow the daily print window and `interval` in `print.prt`.

`basin_psc` rows are sums over all recall objects, so withdrawals (negative flows) offset releases.

Example

The column names of `recall_day.txt`:

```
jday  mon  day  yr  name  type
flo  sed  orgn  sedp  no3  solp  chla  nh3  no2  cbod  dox  san  sil  cla  sag  lag  grv  null
```

A treatment plant entered as a monthly recall object with `flo` = 1,200 for March shows 1,200 on every March row of `recall_day.txt` and 37,200 in the March row of `recall_mon.txt`.

Related files

- Recall inputs: the recall objects and their time series.
- `print.prt`: the `recall` and `basin_psc` switches.
- Hydrograph outputs: the same column layout for water passed between objects; the `hydout` rows of a recall object show where its water goes.
- Output naming convention: all output files.

Sources

- File opening and headers: `header_write.f90`
- Header types: `hyd_header`, `hyd_header_time`, `rec_header_time`, `hyd_header_units`, `hyd_header_units3` in `hydrograph_module.f90`
- Writers: `recall_output.f90`, `basin_recall_output.f90`
- Daily record selection: `case ("recall")` in `command.f90`
- Reading the time series: `recall_read.f90`
- Nutrient removal for withdrawals: `recall_nut.f90`

Column	Fortran variable
hydrograph (`recall`)	`rec_d(irec) = ob(icmd)%hd(1)`, summed into `rec_m`, `rec_y`, `rec_a`
hydrograph (`basin_psc`)	`brec_d` (sum of `rec_d`), `brec_m`, `brec_y`, `brec_a`
`null`	`hyd_output%temp`

Purpose

The carbon output files report the carbon budget of the soil, residue and plants. Two kinds of file are available:

- Flat files with one row per HRU or landscape unit per interval: carbon gains and losses, soil carbon transformations and plant carbon.
- Per-layer files with one row per HRU per interval and one column per soil layer for each variable: soil carbon totals, carbon pools, N and P in the pools, carbon and N fluxes, process drivers, pool dynamics and soil properties.

Every family has a row in `print.prt` with the usual daily, monthly, yearly and average annual flags. The rows are recognised only when `print.prt` uses object labels (`use_obj_lbls = y`); in the fixed-order layout there is no place for them. A row name that SWAT+ does not know stops the run with an error.

Most columns are filled by the dynamic carbon model, which runs when `carbon = 2` in `codes.bsn`. Four per-layer families (`hru_cb_cpool`, `hru_cb_flux`, `hru_cb_drv`, `hru_cb_dyn`) are written only with `carbon = 2`. The other families are written for any setting, but with `carbon = 0` many of their columns stay at zero. See the carbon model theory for the pools and transformations.

Files and switches

`print.prt` row	Files	Rows	Layers
`hru_cb_gl`	`hru_carb_gl_{day,mon,yr,aa}`	HRU	no
`hru_cb_trf`	`hru_scf_{day,mon,yr,aa}`	HRU	no
`hru_cb_lyr`	`hru_cbn_lyr_{day,mon,yr,aa}`	HRU	yes
`hru_cb_cpool`	`hru_cpool_stat_{day,mon,yr,aa}`	HRU	yes, `carbon = 2` only
`hru_cb_npool`	`hru_n_p_pool_stat_{day,mon,yr,aa}`	HRU	yes

print.prt row	Files	Rows	Layers
hru_cb_plt	hru_plc_stat_{day,mon,yr,aa}	HRU	no
hru_cb_flux	hru_cflux_stat_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_drv	hru_carb_drv_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_dyn	hru_carb_dyn_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_snap	hru_soil_snap_{day,mon,yr} and hru_soil_snap_tot	HRU	yes
lsu_cb_gl	lsu_carb_gl_{day,mon,yr,aa}	landscape unit	no
lsu_cb_trf	lsu_scf_{day,mon,yr,aa}	landscape unit	no
lsu_cb_plt	lsu_plc_stat_{day,mon,yr,aa}	landscape unit	no

Each file is written as .txt, plus .csv when csvout = y. The hru_cb_snap row writes _tot in place of _aa; see Soil snapshot. The landscape unit files are written only when the model defines landscape units in ls_unit.def.

Example print.prt rows (labelled layout):

objects	daily	monthly	yearly	avann
hru_cb_gl	n	y	y	y
hru_cb_lyr	n	n	y	y
hru_cb_snap	n	n	n	y

Value types and aggregation

The flat gain and loss files (hru_carb_gl, hru_scf, lsu_carb_gl, lsu_scf) report fluxes. Monthly and yearly files are sums over the interval, and the average annual file divides the run total by the number of printed years.

All other files report a state at the moment the row is written: the monthly file gives the value on the last day of the month, the yearly file the value on the last day of the year, and the average annual file the value at the end of the run. They are not averages. This also applies to the per-layer flux file hru_cflux_stat: its daily, monthly and yearly rows give the flux of the last day of the interval in kg/ha/day, and only the average annual file gives the run total divided by the number of printed years, in kg/ha/yr.

Flat HRU files

Carbon gains and losses

hru_carb_gl_* (row hru_cb_gl). One row per HRU. After the seven leading columns (jday, mon, day, yr, unit, gis_id, name) come 21 columns in kg C/ha, in three groups.

Soil:

Column	Description
sed_c	carbon removed with eroded sediment
surq_c	dissolved carbon in surface runoff
latq_c	dissolved organic carbon in lateral flow, all layers
perc_c	dissolved carbon in percolation below the profile
res_decay	carbon added to the soil by residue decay
man_app_c	carbon added with applied manure
man_graze_c	carbon added with manure from grazing animals
rsp_c	CO_2-C from soil respiration, whole profile
soil_emit_c	CO_2-C from burning soil carbon

Residue:

Column	Description
plant_surf_c	carbon added to surface residue by leaf drop and kill
plant_root_c	carbon added to soil residue by root death
rsd_surfdecay_c	carbon lost from surface residue by decay
rsd_rootdecay_c	carbon lost from root and incorporated residue by decay
harv_stov_c	carbon removed by residue harvest
rsd_emit_c	CO_2-C from burning surface residue

Plant:

Column	Description
npp_c	plant carbon growth (net primary production)
harv_abgr_c	carbon removed by grain and biomass harvest
harv_root_c	carbon removed by tuber (root) harvest
drop_c	carbon moved to residue by leaf drop and kill
grazeat_c	carbon eaten by grazing animals
plant_emit_c	CO_2-C from burning plants

Soil carbon transformations

hru_scf_* (row hru_cb_trf). One row per HRU with 13 columns in kg C/ha: meta_micr, str_micr, str_hs, co2_meta, co2_str, micr_hs, micr_hp, hs_micr, hs_hp, hp_micr, co2_micr, co2_hs, co2_hp. Each name gives the source and destination pool: meta metabolic litter, str structural litter, micr microbial biomass, hs slow humus, hp passive humus, and co2 carbon lost as CO_2.

In 62.0.1 these columns are reset to zero each day and never filled, so every value in hru_scf_* and lsu_scf_* is zero. For the same flows by layer, use hru_cflux_stat (cfmets1 for meta_micr, co2fs1 for co2_micr, and so on).

Plant carbon

hru_plc_stat_* (row hru_cb_plt). One row per HRU with seven columns in kg C/ha:

Column	Description
total_c	carbon in all plant parts
ab_gr_c	carbon in above-ground biomass
leaf_c	carbon in leaves
stem_c	carbon in stems
seed_c	carbon in seeds
root_c	carbon in roots
surf_rsd_c	carbon in residue

Per-layer HRU files

Layout

Each row starts with the seven leading columns, then a depth block depth_lyr1 to depth_lyrN giving the depth to the bottom of each soil layer in mm, then one block of N columns for each variable, named <variable>_lyr1 to <variable>_lyrN. There is no units line: line 1 is the project name and version, line 2 the column header, and the data start on line 3.

N is the same for every HRU. By default it is the largest number of soil layers among all HRUs, so the deepest profile is written in full. The optional file carbon_layers.prt sets N directly. In an HRU with fewer layers than N, the extra columns contain -99.

Soil carbon by layer

`hru_cbn_lyr_*` (row `hru_cb_lyr`). Line 1 ends with the note `total soil carbon (Mg/ha) by layer depth in mm`. The value columns are in Mg C/ha:

Columns	Description
`depth_lyr1` … `depth_lyrN`	depth to the bottom of each layer (mm)
`tot_300_sum`	total soil organic carbon in the top 300 mm
`tot_lyr1` … `tot_lyrN`	total soil organic carbon in each layer
`seq_300_sum`	sequestered (humus and microbial) carbon in the top 300 mm
`seq_lyr1` … `seq_lyrN`	sequestered carbon in each layer

For the 300 mm sums, the layer that crosses 300 mm is counted in proportion to its thickness above 300 mm. With `carbon = 0`, both totals are the sum of the active humus, stable humus and microbial carbon.

Carbon pools

`hru_cpool_stat_*` (row `hru_cb_cpool`, `carbon = 2` only). Ten variables per layer:

Variable	Units	Description
`residue_c`	kg C/ha	residue carbon in the layer
`structural_c`	kg C/ha	structural litter
`metabolic_c`	kg C/ha	metabolic litter
`hs_c`	kg C/ha	slow humus
`hp_c`	kg C/ha	passive humus
`microbial_c`	kg C/ha	microbial biomass
`lignin_c`	kg C/ha	lignin part of structural litter
`nonlignin_c`	kg C/ha	non-lignin part of structural litter
`root_mass`	kg/ha	root mass in the layer
`soil_water`	mm	water in the layer

N and P in the pools

`hru_n_p_pool_stat_*` (row `hru_cb_npool`). Eighteen variables per layer, in kg N/ha and kg P/ha: `tot_pool_n`, `residue_n`, `structural_n`, `metabolic_n`, `hs_n`, `hp_n`, `microbial_n`, `lignin_n`, `nonlignin_n`, then the same nine with `_p`. `tot_pool_n` and `tot_pool_p` add residue, structural and metabolic litter, slow and passive humus, microbial biomass, the dissolved organic pool and manure.

Carbon and N fluxes

`hru_cflux_stat_*` (row `hru_cb_flux`, `carbon = 2` only). Thirty-seven variables per layer. In the names, `s1` is microbial biomass, `s2` slow humus, `s3` passive humus, `met` metabolic litter and `str` structural litter, so `cfs1s2` is the carbon flow from microbial biomass to slow humus.

Variables	Units	Description
`cfmets1`, `cfstrs1`, `cfstrs2`	kg C/ha/day	carbon moved from litter to microbial biomass or slow humus
`efmets1`, `efstrs1`, `efstrs2`	kg N/ha/day	nitrogen moved with those flows
`immmets1`, `immstrs1`, `immstrs2`	kg N/ha/day	N immobilised by the litter flows
`mnrmets1`, `mnrstrs1`, `mnrstrs2`	kg N/ha/day	N mineralised by the litter flows
`co2fmet`, `co2fstr`	kg C/ha/day	CO_2-C from metabolic and structural litter
`cfs1s2`, `cfs1s3`, `cfs2s1`, `cfs2s3`, `cfs3s1`	kg C/ha/day	carbon moved between the humus and microbial pools
`efs1s2`, `efs1s3`, `efs2s1`, `efs2s3`, `efs3s1`	kg N/ha/day	nitrogen moved with those flows
`imms1s2`, `imms1s3`, `imms2s1`, `imms2s3`, `imms3s1`	kg N/ha/day	N immobilised by those flows
`mnrs1s2`, `mnrs1s3`, `mnrs2s1`, `mnrs2s3`, `mnrs3s1`	kg N/ha/day	N mineralised by those flows
`co2fs1`, `co2fs2`, `co2fs3`	kg C/ha/day	CO_2-C from microbial biomass, slow humus and passive humus

The average annual file gives yearly totals (kg/ha/yr); see Value types and aggregation.

Process drivers

`hru_carb_drv_*` (row `hru_cb_drv`, `carbon = 2` only). Fourteen variables per layer, all for the day the row is written:

Variable	Units	Description
`sut`	fraction	soil water control on decomposition
`tillagef`	-	tillage factor of the layer
`cons_bmix`	-	biological mixing coefficient of the layer
`tillagef_biomix`	-	part of the tillage factor from biological mixing
`tillagef_tillmix`	-	part of the tillage factor from tillage operations
`till_eff`	fraction	tillage effect on decomposition
`cdg`	fraction	soil temperature control on decomposition
`ox`	fraction	oxygen control on decomposition
`cs`	fraction	combined control on decomposition
`no3`	kg N/ha	nitrate available to the transformations
`nh4`	kg N/ha	ammonium available to the transformations
`co2_resp`	kg C/ha	CO_2-C respired in the layer
`soil_temp`	°C	soil temperature
`emix`	fraction	fraction of the layer mixed by tillage or biological mixing

Pool dynamics

`hru_carb_dyn_*` (row `hru_cb_dyn`, `carbon = 2` only). Twenty-one variables per layer:

Variables	Units	Description
`asp`, `abp`	fraction	share of decomposed slow humus and of decomposed microbial biomass sent to passive humus
`abco2`, `a1co2`, `asco2`, `apco2`	fraction	share of decomposed microbial biomass, metabolic and passive pools, slow humus and passive humus lost as CO_2
`ncbm`, `nchp`, `nchs`	ratio	N:C ratio of microbial biomass, passive humus and slow humus
`bmctp`, `bmntp`	kg/ha/day	potential C and N transformation of microbial biomass
`hsctp`, `hsntp`	kg/ha/day	potential C and N transformation of slow humus
`hpctp`, `hpntp`	kg/ha/day	potential C and N transformation of passive humus
`lmctp`, `lmntp`	kg/ha/day	potential C and N transformation of metabolic litter
`lsctp`, `lslctp`, `lslnctp`, `lsntp`	kg/ha/day	potential transformation of structural litter: C, lignin C, non-lignin C and N

Soil snapshot

`hru_soil_snap_*` (row `hru_cb_snap`). Thirteen soil properties per layer:

Variable	Units	Description
`bd`	Mg/m^3	bulk density
`awc`	mm/mm	available water capacity
`soil_k`	mm/h	saturated hydraulic conductivity
`tot_c`	kg C/ha	total soil organic carbon
`clay`, `silt`, `sand`, `rock`	%	particle size fractions and rock fragments
`alb`	fraction	albedo
`usle_k`	-	USLE soil erodibility
`ec`	dS/m	electrical conductivity
`caco3`	%	calcium carbonate content
`ph`	-	soil pH

The daily, monthly and yearly flags write `hru_soil_snap_day`, `_mon` and `_yr`. The average annual flag writes `hru_soil_snap_tot` instead, which has one row per HRU at the start of the run (marked `jday = 0`) and one at the end. Both rows give `tot_c` in kg C/ha, so the change in soil carbon in layer K over the run is `tot_c_lyrK` on the end row minus `tot_c_lyrK` on the start row. The start rows are written only when `carbon = 2`.

Repeated rows

In 62.0.1 the files of the `hru_cb_lyr`, `hru_cb_cpool`, `hru_cb_npool`, `hru_cb_plt`, `hru_cb_flux`, `hru_cb_drv` and `hru_cb_dyn` rows, and the daily, monthly and yearly files of `hru_cb_snap`, repeat their block of HRU rows once per HRU. In a model with 12 HRUs, every interval has 144 rows: the 12 HRU rows written 12 times with the same values. Keep the first occurrence of each `unit` within an interval when reading these files. `hru_soil_snap_tot`, `hru_carb_gl`, `hru_scf` and the landscape unit files are not affected.

Landscape unit files

The landscape unit files are area-weighted averages of the HRU values, each HRU weighted by its fraction of the landscape unit. Per-layer files have no landscape unit version because the HRUs in a unit can have different soil profiles.

File	Columns
`lsu_carb_gl_*`	the 21 columns of `hru_carb_gl`
`lsu_scf_*`	the 13 columns of `hru_scf` (all zero in 62.0.1)
`lsu_plc_stat_*`	one column, `lsu_plt_c`: carbon in all plant parts (kg C/ha)

`lsu_plt_c` is a state value: the monthly, yearly and average annual files give the value on the last day of the interval.

Layer count file

`carbon_layers.prt` is optional and is not listed in `file.cio`. When it is present in the run folder, it fixes N, the number of layer columns in every per-layer file. Use it to keep the same number of columns across runs, or to write only the top layers.

- Line 1: title, skipped.
- Line 2: header, skipped.
- Line 3: the number of layers, a whole number of 1 or more.

```
carbon_layers.prt: layers to expand in per-layer carbon output
n_layers
10
```

With `10`, every per-layer file has ten columns per variable, whatever the soils. A value below 1 is rejected with a message in `diagnostics.out`, and N then defaults to the largest layer count among the HRUs.

Legacy CSU carbon outputs

An older set of carbon files, kept for existing CSU workflows, runs beside the files above. SWAT+ Editor does not write the rows that switch them on; they are added to `print.prt` by hand. The code marks them for removal in revision 63.

`print.prt` row and flag	Files
`hru_cb` with `y` or `l` in the daily, monthly or yearly column	`hru_cbn_lyr.txt`, `hru_seq_lyr.txt`, `hru_n_p_pool_stat.txt`

print.prt row and flag	Files
as above, with `1` in any column	adds `hru_begsim_soil_prop.txt` and `hru_endsim_soil_prop.txt` (the start rows need `carbon = 2`)
as above, with `carbon = 2`	adds `hru_plc_stat.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt`
`hru_cb` with any flag other than `n`	`basin_carbon_all.txt` (header lines only when `hru_nb` has `y` in the average annual column)
`hru_cb_vars` with `y` or `1` in the daily, monthly or yearly column, and `carbon = 2`	`hru_carbvars.txt`, `hru_org_allo_vars.txt`, `hru_org_ratio_vars.txt`, `hru_org_trans_vars.txt`

These files have no interval suffix: one file collects the rows for every interval switched on. Each also gets a `.csv` copy when `csvout = y`. The legacy files contain the same quantities as the files above in a different layout. For example, legacy `hru_cbn_lyr.txt` and `hru_seq_lyr.txt` together contain the values that the new `hru_cbn_lyr_yr.txt` gives in one table.

The legacy files do not use the layer count file. The `cbn_diagnostics` setting from the old `carb_coefs.cbn` no longer exists; the `1` flag on the `hru_cb` row takes its place.

Example

Header and first row of `hru_carb_gl_yr.txt` from a two-year run of the `Ames_sub1` reference dataset with 62.0.1 (`carbon = 2`), shortened to the first eleven value columns:

```
      jday        mon        day         yr            unit              gis_id     name

       365         12         31       1975               1                   1 hru0001          0.702
```

Start of the first data row of `hru_cbn_lyr_yr.txt` from the same run, as values per column. Every soil in this dataset has 16 layers, so N = 16 and no column contains `-99`:

Column	Value
`unit`	1
`depth_lyr1` to `depth_lyr16`	10, 25, 50, 100, 150, 200, 300, 450, 600, 750, 900, 1050, 1200, 1500, 1800, 2000
`tot_300_sum`	44.06
`tot_lyr1`, `tot_lyr2`, `tot_lyr3`	2.28, 2.55, 4.24

The 12 HRU rows for 1975 appear 12 times each in this file (see Repeated rows).

Related files

- `print.prt`: output switches.
- `codes.bsn`: the `carbon` switch.
- `carbon.bsn` and `carbon_lyr.bsn`: carbon model inputs.
- Carbon model theory: pools and transformations.
- Naming convention: file suffixes and leading columns.

Sources

- `print.prt` rows: `basin_print_codes_read.f90`

- File opening and headers: output_landscape_init.f90, header types output_carb_gl_header and output_hscf_header in output_landscape_module.f90, variable lists and header and row helpers in carbon_module.f90
- Flat HRU files: hru_carbon_output.f90 (units 4520 to 4527 and 4550 to 4557)
- Per-layer files: soil_nutcarb_write.f90 and soil_carbvar_write.f90 (units 4530 to 4605), called from command.f90 and, for the end-of-run snapshot, hru_output.f90
- Landscape unit files: lsu_carbon_output.f90 (units 4750 to 4773), landscape unit count from lsu_read_elements.f90
- Layer count file: carbon_layers_read.f90
- Legacy files: carbon_legacy_module.f90 (units 8348 to 8387)

print.prt row	Flag variable
hru_cb_gl, hru_cb_trf, hru_cb_lyr, hru_cb_cpool, hru_cb_npool	pco%cb_gl_hru, pco%cb_trf_hru, pco%cb_lyr_hru, pco%cb_cpool_hru, pco%cb_npool_hru
hru_cb_plt, hru_cb_flux, hru_cb_drv, hru_cb_dyn, hru_cb_snap	pco%cb_plt_hru, pco%cb_flux_hru, pco%cb_drv_hru, pco%cb_dyn_hru, pco%cb_snap_hru
lsu_cb_gl, lsu_cb_trf, lsu_cb_plt	pco%cb_gl_lsu, pco%cb_trf_lsu, pco%cb_plt_lsu
hru_cb, hru_cb_vars (legacy)	pco%cb_hru, pco%cb_vars_hru

Output columns	Fortran source
hru_carb_gl	hsc_d, hrc_d, hpc_d (carbon_soil_gain_losses, carbon_residue_gain_losses, carbon_plant_gain_losses); res_decay is rsd_decay_c, man_graze_c is man_graz_c, the three *_emit_c columns are each type's emit_c
hru_scf	hscf_d (carbon_soil_transformations)
tot_lyrK, seq_lyrK, tot_300_sum, seq_300_sum	soil1%tot(k)%c, soil1%seq(k)%c, soil1%tot_300_c, soil1%seq_tot_300_c, all divided by 1000
hru_cflux_stat	soil1%org_flx_lr(k); _aa uses soil1%org_flx_cum_lr(k) divided by yrs_prt
hru_carb_drv	soil1%org_con_lr(k), soil%ly(k)%tillagef*, soil%ly(k)%bmix, soil%phys(k)%tmp, soil1%emix(k)
hru_carb_dyn	soil1%org_allo_lr(k), soil1%org_ratio_lr(k), soil1%org_tran_lr(k)
lsu_plt_c	sum of pl_mass%tot_com%c * lsu_elem%ru_frac

Purpose

The gwflow module simulates groundwater on a grid of cells and replaces the one-dimensional aquifers. It runs when gw_flow = 1 in codes.bsn and writes the output files described here. The input files are described on the gwflow input page.

The files fall into these groups:

- basin and cell water balance (gwflow_basin_wb_*, gwflow_cell_wb_*)
- heat and solute balances (gwflow_basin_heat_*, gwflow_basin_sol_*)
- observation wells (gwflow_obs_*)
- HRU pumping (gwflow_hru_pump_*)
- diagnostic files for canals, ponds, tiles and channels
- channel hydrograph separation (gwflow_chan_hydsep_day.txt)
- the cell definition and the run log, which are always written

Output switches

The write_day, write_mon, write_yr and write_aa flags in codes.gw (1 on, 0 off) switch the water balance files. The heat and solute balance files follow the same flags, and the hydrograph separation file follows write_day. Observation well, pumping and diagnostic output is on by default.

print.prt can override these settings with six rows, each with the usual daily, monthly, yearly and average annual flags. The rows are recognised only when print.prt uses object labels (use_obj_lbls = y). A row that is present replaces the default; a row that is absent leaves it as it is. All overrides are applied before any gwflow output file is opened, so a file that is switched off is not created.

print.prt row	Effect
gwflow_wb	each flag switches the water balance file for that interval; heat, solute and hydrograph separation files follow it unless their rows are present
gwflow_heat	each flag switches the heat balance file for that interval (there is no monthly heat file)
gwflow_solute	each flag switches the solute balance files for that interval
gwflow_obs	each flag switches the observation well file for that interval
gwflow_pump	any flag set to y switches on all HRU pumping files
gwflow_flux	the daily or yearly flag switches on the canal, pond, tile and channel diagnostic files

Example rows that keep only the monthly water balance and the daily observation wells:

objects	daily	monthly	yearly	avann
gwflow_wb	n	y	n	n
gwflow_obs	y	n	n	n
gwflow_pump	n	n	n	n
gwflow_flux	n	n	n	n

Basin water balance

gwflow_basin_wb_day.txt, _mon.txt, _yr.txt and _aa.txt give the water balance of the whole grid. Volumes are summed over all active cells and divided by the watershed area, so they are in mm over the watershed. Line 1 is a title, line 2 the column header and line 3 the units. The seven leading columns are jday, mon, day, yr, unit, gis_id and name (the project name).

The daily file has these columns after the leading ones:

Column	Units	Description
ts	day	gwflow time step
vbef	mm	groundwater volume at the start of the day
vaft	mm	groundwater volume at the end of the day
recharge	mm	recharge from the soil profile
gw_et	mm	groundwater evapotranspiration
gw_sw	mm	groundwater discharge to channels
sw_gw	mm	channel seepage to groundwater
sat_excess	mm	saturation excess flow to the land surface
soil	mm	groundwater moved into the soil profile
lateral	mm	net lateral flow between cells
boundary	mm	exchange across the grid boundary
pump_allo	mm	pumping for water allocation (irrigation)
pump_ext	mm	pumping specified in pumpex.gw
tile	mm	tile drain outflow
reservoir	mm	exchange with reservoirs
wetland	mm	exchange with wetlands
canal	mm	canal seepage
floodplain	mm	exchange with floodplains

Column	Units	Description
pond	mm	recharge pond seepage
phytorem	mm	phreatophyte transpiration
error	%	volume balance error
sat_frac	fraction	fraction of active cells with saturation excess flow that day
wt_depth	m	average depth to the water table (see the note below)
pump_def	mm	allocation pumping demand that could not be met

The monthly, yearly and average annual files replace ts, vbef and vaft with dvol (change in storage, mm) and drop error, sat_frac and wt_depth. They end with pump_def after phytorem.

The daily wt_depth is the average over the active cells. In the average annual file, dvol is the total change in storage over the run divided by the number of years, and every flux is the run total divided by the number of years.

Cell water balance

gwflow_cell_wb_day.txt, _mon.txt, _yr.txt and _aa.txt have one row per active cell per interval. Lines 1 and 2 are notes on units; line 3 is the column header.

Column	Units	Description
jday, mon, day, yr	-	date of the row
unit	-	cell number in the gwflow grid
gis_id	-	cell ID from the gis_id column of cells.gw (the gwflow cell shapefile ID)
name	-	cell name from cells.gw, such as cell0001 (first 12 characters)
head	m	groundwater head
wt_depth	m	land surface elevation of the cell minus the head
recharge	m^3/day	recharge
gw_et	m^3/day	groundwater evapotranspiration
gw_sw	m^3/day	discharge to channels
sw_gw	m^3/day	channel seepage
sat_excess	m^3/day	saturation excess flow
soil	m^3/day	transfer to the soil profile
lateral	m^3/day	net lateral flow
pump_allo	m^3/day	allocation pumping
pump_ext	m^3/day	specified pumping
tile	m^3/day	tile drain outflow
reservoir	m^3/day	reservoir exchange
wetland	m^3/day	wetland exchange
floodplain	m^3/day	floodplain exchange
canal	m^3/day	canal seepage
pond	m^3/day	recharge pond seepage
phytorem	m^3/day	phreatophyte transpiration

The monthly, yearly and average annual files give the average daily rate over the interval, so all fluxes stay in m^3/day.

wt_depth is measured from the cell's land surface elevation, which on a coarse grid represents the valley and not the streambed. Cells along channels can therefore show a deep water table while the head is close to the channel level. This comes from the grid resolution; the head itself is not affected.

Heat balance

When heat transport is on (heat = 1 in codes.gw), gwflow_basin_heat_day.txt, _yr.txt and _aa.txt are written. They follow the daily, yearly and average annual water balance switches, or the gwflow_heat row of print.prt when it is present; there is no monthly heat file. Values are in MJ.

The daily file has `ts`, `hbef` and `haft` (heat stored at the start and end of the day), then `recharge`, `gw_et`, `gw_sw`, `sw_gw`, `sat_excess`, `soil`, `lateral`, `dispersion`, `boundary`, `pump_allo`, `pump_ext`, `tile`, `reservoir`, `wetland`, `canal`, `floodplain`, `pond`, then `error` (%) and `tavg` (average groundwater temperature, °C). The yearly and average annual files start with `hdel` (change in heat stored) and end at `pond`. ## Solute balances {#output-reference-gwflow-solute-balances}

When solute transport is on (`solute = 1` in codes.gw), one file per solute and interval is written, named `gwflow_basin_sol_<solute>_<interval>.txt`. Nitrate (`no3`) and phosphorus (`p`) are always included; with salts in the model, `so4`, `ca`, `mg`, `na`, `k`, `cl`, `co3` and `hco3` are added; with constituents, `seo4` and `seo3`. Values are in kg. The interval files follow the water balance switches, or the `gwflow_solute` row of print.prt when it is present.

The daily file has `ts`, `mbef` and `maft` (solute mass at the start and end of the day), then `recharge`, `gw_sw`, `sw_gw`, `sat_excess`, `soil`, `advection`, `dispersion`, `react_in`, `react_out`, `mineral`, `sorption`, `pump_allo`, `pump_ext`, `tile`, `reservoir`, `wetland`, `canal`, `floodplain`, `pond` and `error` (%). The monthly, yearly and average annual files start with `delm` (change in mass) and end at `pond`.

Observation wells

Observation cells are listed as `observation_cell` lines in outputs.gw. The files `gwflow_obs_day.txt`, `_mon.txt`, `_yr.txt` and `_aa.txt` have one row per observation cell:

Column	Units	Description
`unit`	-	observation well number, in the order listed in outputs.gw
`gis_id`	-	cell number of the observation cell
`name`	-	`obs_0001`, `obs_0002`, …
`head`	m	groundwater head
`wt_depth`	m	land surface elevation minus head
`temperature`	°C	groundwater temperature; `-99` when heat transport is off
`no3`	mg/L	nitrate concentration; `-99` when solute transport is off
`p`	mg/L	phosphorus concentration; `-99` when solute transport is off

The monthly and yearly files give averages over the interval, and the average annual file the average over the run.

HRU pumping

When pumping for water allocation is simulated, `gwflow_hru_pump_day.txt`, `_mon.txt`, `_yr.txt` and `_aa.txt` report the groundwater pumped for each HRU in the `pump_allo` column. The daily file is in m^3/day and lists only HRUs that pumped that day; the monthly and yearly files give totals in m^3, and the average annual file m^3/yr.

When hru_pump.gw lists HRUs to observe, `gwflow_cell_wb_ppag_obs_day.txt` gives their daily pumping in m^3, one column per listed HRU. It is written under the same switch as the other pumping files.

Diagnostic files

These files are written when the related feature is in use and diagnostic output is on (`gwflow_flux`):

File	Content
`gwflow_canal_wb_day.txt`	daily canal water balance in m^3: `diversion`, `storage`, `pond_xfer`, `seepage`
`gwflow_canal_sol_day.txt`	daily canal solute balance in kg: `div_mass`, `storage`, `pond_mass`, `seep_mass` for each solute
`gwflow_pond_wb_day.txt`	daily recharge pond water balance: `area` (m^2), `storage`, `rain`, `div_added`, `evap`, `recharge`, `div_spec`, `div_unsat` (m^3)
`gwflow_pond_sol_day.txt`	daily pond solute balance: `area`, `storage`, `solute`, `mass`, `div_added`, `recharge` (kg)
`gwflow_pond_mass_day.txt`, `gwflow_pond_conc_day.txt`	daily solute mass (kg) and concentration (g/m^3), one column per pond

File	Content
`gwflow_tile_group_day.txt`	daily tile flow (m^3/s) for each group of tile cells
`gwflow_gwsw_group_day.txt`	daily groundwater and channel exchange (m^3) for each channel cell group in `sw_group.gw`
`gwflow_chan_obs_flow_day.txt`	daily groundwater and channel exchange (m^3) at the channel cells flagged in `chancell.gw`
`gwflow_chan_obs_no3_day.txt`	the same for nitrate (g), when solute transport is on

When `gwflow.wbgroups` is present, one daily water balance file per cell group is also written, named `gwflow_group_wb_day_` followed by the group number as two characters (a space and the digit for groups 1 to 9). Its columns follow the daily basin file, in m^3.

Hydrograph separation

`gwflow_chan_hydsep_day.txt` is written when daily water balance output is on. It gives, for each channel and day, the flow from each source in m^3/s: `surf_runoff`, `lateral`, `gw_discharge`, `stream_seep`, `satex_gw`, `satex_sw` and `tile_drain`.

Files always written

File	Content
`gwflow_record`	run log: which input files were found, which features are active, which `print.prt` rows changed the output switches, and error messages from reading
`gwflow_cell_definition.txt`	one line per cell: `cell_id`, `row`, `col`, `x_coord`, `y_coord`, `zone`, `status`, `area`

When `transit.gw` is present on a structured grid, `gwflow_transit_cell`, `gwflow_transit_chan` and (with tiles) `gwflow_transit_tile` give travel times in days.

The `head_output_time` lines in `outputs.gw` are still read but no longer produce output; cell heads are in the cell water balance files.

Related files

- gwflow input files
- `codes.bsn`: `gw_flow` switch
- `print.prt`: output rows
- Aquifer theory

Sources

- Output file setup and basin, cell, heat, solute and observation writers: `gwflow_output.f90` (`gwflow_output_init`, `gwflow_output_day`, the monthly, yearly and average annual routines, `gwflow_write_celldef`)
- `codes.gw` flags, `print.prt` overrides, pumping, diagnostic and hydrograph separation file setup: `gwflow_read.f90`
- Flags and unit numbers: `gwflow_module.f90`
- Daily simulation, pumping output and the call to the average annual output: `gwflow_simulate.f90`; pond and canal diagnostics in `gwflow_pond.f90` and `gwflow_canal_div.f90`
- `print.prt` rows: `basin_print_codes_read.f90`; `gwflow_record` opened in `basin_read_objs.f90`

Switch	Fortran variable
`write_day`, `write_mon`, `write_yr`, `write_aa` in `codes.gw`, or the `gwflow_wb` row	`gwflag_day`, `gwflag_mon`, `gwflag_yr`, `gwflag_aa`
`gwflow_heat` row (default: water balance flags)	`gwflag_heat_day`, `gwflag_heat_yr`, `gwflag_heat_aa`
`gwflow_solute` row (default: water balance flags)	`gwflag_solute_day`, `gwflag_solute_mon`, `gwflag_solute_yr`, `gwflag_solute_aa`
`gwflow_obs` row (default: all on)	`gwflag_obs_day`, `gwflag_obs_mon`, `gwflag_obs_yr`, `gwflag_obs_aa`, and `gwflag_obs` as their maximum
`gwflow_pump`, `gwflow_flux` rows (default: on)	`gwflag_pump`, `gwflag_flux`
hydrograph separation	`hydsep_flag`, set from `gwflag_day`
`heat`, `solute` in `codes.gw`	`gw_heat_flag`, `gw_solute_flag`

Cell file column	Fortran variable
`gis_id`, `name`	`cell_gis_id(i)`, `cell_name(i)` (written with `a12`)
`recharge` … `phytorem`	`gw_hyd_ss(i)%rech`, `gwet`, `gwsw`, `swgw`, `satx`, `soil`, `latl`, `ppag`, `ppex`, `tile`, `resv`, `wetl`, `fpln`, `canl`, `pond`, `phyt`

Purpose

SWAT+ writes the files on this page on every run; `print.prt` cannot switch them off. They show whether the run finished, which inputs SWAT+ could not match, the soil and HRU settings it used, and which output files it wrote. Check them before using any other output.

File	Content
`simulation.out`	run log: version banner, weather files read, one line per simulated day, completion message
`diagnostics.out`	input problems found while reading: names not found, files missing, values rejected
`checker.out`	one row per HRU with soil and hydrology settings after setup
`files_out.out`	list of output files opened in this run
`success.fin`	written only at the end of a run that finished
`area_calc.out`	drainage areas of routed objects, as given and as calculated
`erosion.out`	header lines only in 62.0.1
`erosion.txt`	empty in 62.0.1
`paddy_test.csv`	daily paddy rice water balance; column header only unless the model has paddy wetlands

With the gwflow module on, `gwflow_record` is also always written; see gwflow outputs.

simulation.out

The run log. It contains, in order:

- a banner with the revision, the compiler and its version, the build date and the operating system
- the date and time of the run
- one line for each weather input read (`reading from precipitation file`, `reading from temperature file`, and so on)
- one line per simulated day, giving the simulation name, month, day, year, the year number and the number of years, and the clock time
- at the end, `Execution successfully completed` and the date and time again

Each daily line starts with `Original Simulation` in a normal run. During soft calibration other names appear in that place.

```
                  SWAT+
              Revision 62.0.1
      Soil & Water Assessment Tool
IntelLLVM (2025.0.4), 2026-10-05 00:10:27, Linux
    Program reading . . . executing


 Date of Sim  10/20/2026 Time  12: 9:24
 reading from wgn file                                    Time  12: 9:24
 reading from wx station file                             Time  12: 9:24
 reading from precipitation file                          Time  12: 9:24
 reading from temperature file                            Time  12: 9:24
  ...
 Original Simulation            1   1  1975 Yr    1 of    2 Time  12: 9:24
  ...
 Original Simulation           12  31  1976 Yr    2 of    2 Time  12: 9:35

 Execution successfully completed

  Date of Sim  10/20/2026 Time  12: 9:35
```

If the completion message is missing, the run stopped early. The last daily line shows the last day that finished, so the problem occurred on the next day. Messages printed to the screen when the run stopped are not copied to this file.

diagnostics.out

Line 1 is `DIAGNOSTICS.OUT FILE`. Every later line reports a problem found while reading the inputs. About 60 places in the code write to this file. Most lines say that a name given in one file was not found in the file it points to; SWAT+ then uses a default or skips the setting and continues.

Common messages:

Message	Meaning
`<land use> <schedule> not found in management.sch`	a row of `landuse.lum` names a management schedule that is not in `management.sch`
`<name> not found (snow.sno)`, `(soils.sol)`, `(topography.hyd)`, and similar	an HRU or object points to a record that does not exist; the default record is used
`<file> file not found (tgage)`, `(pgage)`, and similar	a weather station names a climate file that is missing
`... could not be opened`	an input file listed in `file.cio` could not be opened
`... is required when carbon is enabled (codes.bsn carbon = 2)`	a carbon input file is missing while the dynamic carbon model is on
`carbon_layers.prt: n_layers must be >= 1 ...`	the optional layer count file has an invalid value

Example from a run of the `Ames_sub1` reference dataset with 62.0.1:

```
 DIAGNOSTICS.OUT FILE
 agrr_lum                              AGRR_rot                                  not found in management.
 soyb_lum                              SOYB_rot                                  not found in management.
 oats_lum                              OATS_rot                                  not found in management.
 wwht_lum                              WWHT_rot                                  not found in management.
 ames.tem                                        file not found (tgage)
```

A file with only the first line means no problems were found. Read the whole file after each run and decide for each line whether the default SWAT+ used is acceptable.

checker.out

One row per HRU, in HRU order, written after the HRUs are set up. Line 1 is the project name and version, line 2 the column header and line 3 the units; the last unit entry, `0=notile;1=tile;`, explains the `tiledrain` column.

#	Column	Units	Description
1	`sname`	-	soil name used by the HRU (no header label; the name is in the units line)
2	`hydgrp`	-	hydrologic soil group, A to D (no header label)
3	`zmx`	mm	maximum rooting depth of the soil
4	`usle_k`	-	USLE soil erodibility factor
5	`sumfc`	mm	water held at field capacity, whole profile
6	`sumul`	mm	water held at saturation, whole profile
7	`usle_p`	-	USLE support practice factor
8	`usle_ls`	-	USLE slope length and steepness factor
9	`esco`	-	soil evaporation compensation factor
10	`epco`	-	plant uptake compensation factor
11	`cn3_swf`	-	soil water factor for curve number condition III
12	`perco`	-	percolation coefficient
13	`latq_co`	-	lateral flow coefficient
14	`tiledrain`	0 or 1	1 if the HRU has tile drainage

The file has no HRU number or name column. Row K belongs to HRU K in `hru-data.hru`.

Example from a run of `Ames_sub1` with 62.0.1:

```
 demo                      SWAT+ 2026-10-05        MODULAR Rev 2026.62.0.1
                                        zmx       usle_k       sumfc       sumul      usle_p     usle_ls
 sname            hydgrp                (mm)                    (mm)        (mm)
soil_01-h1        B                2000.0000       0.0000    361.5003    616.0899      1.0000      0.3638
soil_02           B                2000.0000       0.0000    361.5004    449.0375      0.7500      0.3638
```

Look for zero, negative or extreme values: here `usle_k` is zero for both soils, so these HRUs produce no USLE erosion. Check whether that is intended.

files_out.out

Line 1 is `files_out.out - OUTPUT FILES WRITTEN`. Each later line names one output file opened in the run, after a tag for its type, such as `CHK`, `HRU`, `LSU`, `BASIN`, `BASIN_AQUIFER`, `BASIN_CROP_YLD`, `CROP`, `RES` or `DTBL`. `.csv` copies are listed separately.

```
files_out.out - OUTPUT FILES WRITTEN
CHK                       checker.out
HRU                       hru_carb_gl_mon.txt
HRU                       hru_carb_gl_mon.csv
```

Use it to confirm that every file you asked for in `print.prt` was opened. A file missing from the list was not switched on, or its object type is not in the model (for example, reservoir files in a model without reservoirs). Some files are not listed even when they are written, including the gwflow files and `reservoir_mon.csv`.

success.fin

Written at the end of a run that finished. It contains a blank line and `Execution successfully completed`. Calibration tools and scripts test for this file to tell whether a run finished. Delete it before starting a run if you rely on it, because SWAT+ does not remove an old copy at the start.

area_calc.out

Drainage areas of every routed object except HRUs and routing units: the object number, type and number within its type, the drainage area given in the connection file and the area SWAT+ calculated from the routing, followed by the same areas for each incoming object. Use it to check that the routing delivers the expected area to each channel and outlet. The file is empty when the model has only HRUs and routing units.

Other files

`erosion.out` contains three header lines and no data in 62.0.1. `erosion.txt` is opened at the start of each run and left empty.

`paddy_test.csv` is a temporary paddy rice output. It has a header line, and one line per day for each HRU that has a paddy wetland with water: date, HRU, precipitation, irrigation, seepage, PET, soil evaporation, plant transpiration, weir height, water depth, weir outflow, soil water, sediment concentration and yield, nitrate concentration and yield, LAI, salt concentration and soil water as a percentage of field capacity.

Checks after a run

1. `simulation.out` ends with `Execution successfully completed`, and `success.fin` exists.
2. `diagnostics.out` has only its first line, or every remaining line is understood and accepted.
3. `checker.out` has plausible values for every HRU, and `tiledrain` matches the design.
4. `files_out.out` lists every file requested in `print.prt`.

Related files

- Naming convention: file names and suffixes.
- Post-processing helpers: includes a parser for `checker.out`.
- `print.prt`: switches for the other output files.

Sources

- `simulation.out`, `success.fin`, `erosion.txt`, `paddy_test.csv`: `main.f90.in` (units 9003, 107, 888, 100100); weather lines in `proc_date_time.f90`; daily lines in `time_control.f90`; paddy rows in `hru_control.f90`
- `files_out.out`, `diagnostics.out`, `area_calc.out`: opened in `proc_bsn.f90` (units 9000, 9001, 9004); `area_calc.out` rows in `hyd_connect.f90`
- `checker.out`, `erosion.out`: `proc_hru.f90` (units 4000, 4001); header types `output_checker_header` and `output_checker_unit` in `hydrograph_module.f90`

`checker.out` column	Fortran variable
`sname`, `hydgrp`, `zmx`, `usle_k`, `sumfc`, `sumul`	`soil(j)%snam`, `%hydgrp`, `%zmx`, `%usle_k`, `%sumfc`, `%sumul`
`usle_p`, `usle_ls`	`hru(j)%lumv%usle_p`, `hru(j)%lumv%usle_ls`
`esco`, `epco`, `cn3_swf`, `perco`, `latq_co`	`hru(j)%hyd%esco`, `%epco`, `%cn3_swf`, `%perco`, `%latq_co`
`tiledrain`	`hru(j)%tiledrain`

Purpose

The `test` folder of the SWAT+ source has two Python scripts. They are not installed with the model; run them from a copy of the source or copy them next to your project.

- `spcheck.py` runs SWAT+ on a scenario and compares the numbers in two sets of output files. The build uses it for its regression test.
- `swat_output_to_csv_files.py` reads SWAT+ output files into pandas tables and can save them as tab-separated files.

spcheck.py

`spcheck.py` needs Python 3.10 or later and no extra packages. It has three commands.

run

`python spcheck.py run <executable> <scenario>`

Run from a folder that contains `build/`, `data/` and `runs/`. The command copies `data/<scenario>` to a new folder `runs/<scenario>/<NN>#<date and time>#<executable>`, where `NN` is the next two-digit run number (`01`, `02`, …), and runs `build/<executable>` there. If SWAT+ ends with an error, the run folder gets the suffix `-FAILED`.

compare

`python spcheck.py compare <scenario> <a> <b> [--abserr 1e-8] [--relerr 0.05] [--nlines N] [--nerrorlines N`

Compares two earlier runs of a scenario, given by their run numbers `a` and `b`. The files to compare are listed in `data/<scenario>/.testfiles.txt`, one name per line; lines starting with `#` are skipped. A failed run cannot be compared.

ctest

`python spcheck.py ctest <model_path> <data_dir> <tmp_dir> [--abserr 1e-8] [--relerr 0.05] [--nlines N] [--`

Copies the whole folder `<data_dir>`, including its output files, to `<tmp_dir>`, runs `<model_path>` there, and compares the files listed in `<data_dir>/.testfiles.txt` between the original and the new run. It exits with code 1 when it finds differences. Because the original outputs are copied too, a listed file that the new run does not write is compared with the copied original and passes.

Comparison rule

Both files are read line by line. On each line, only tokens that are decimal numbers are compared, by position; whole numbers (dates, object numbers) and text are skipped, and lines containing the version banner `MODULAR` are skipped. Two values f_1 (first run) and f_2 (second run) differ when

$$|f_1 - f_2| \geq e_{abs} + e_{rel}\,|f_2|$$

with e_{abs} = `--abserr` (default 10^{-8}) and e_{rel} = `--relerr` (default 0.05). For each differing line the script prints both lines, the field numbers, the two values and their absolute and relative error, and at the end the total number of differences.

The script assumes both files have the same layout. It stops at the end of the shorter file without reporting a difference in length. A line whose two versions have different numbers of decimal values is not reported, and it lowers the difference count by one. `--nlines` limits the number of lines checked per file, and `--nerrorlines` the number of differing lines reported per file.

Regression test in the build

`CMakeLists.txt` registers one test, `Ames_sub1`, which calls `spcheck.py ctest` with `--relerr 0.01` and `--abserr 1e-8`. The test reads its scenario from `<source>/data/Ames_sub1`; the reference data in the 62.0.1 source is in `refdata/Ames_sub1`. The `.testfiles.txt` of `Ames_sub1` lists 42 files; six of them (`hru_nut_carb_gl_aa.txt`, `hru_soilcarb_aa.txt`, `hru_rescarb_aa.txt`, `hru_plcarb_aa.txt`, `hru_soilc_stat.txt`, `hru_resc_stat.txt`) are no longer written by 62.0.1 and pass only because of the copy described above.

swat_output_to_csv_files.py

The script maps output file names to reading instructions for `pandas.read_csv` and loads each file into a table.

Use

```
cd <folder with SWAT+ outputs>
python /path/to/swat_output_to_csv_files.py
```

The script reads `files_out.out` from the current folder, then reads every file listed there, except `mgt_out.txt` and `yield.out`, and writes each one to `<name>.csv` in the same folder (for example `hru_wb_aa.txt.csv`). The `.csv` files are tab-separated.

From Python, one file can be read without writing anything:

```
import swat_output_to_csv_files as s
df = s.read_output("basin_wb_aa.txt", s.spec_dict, write_csv=False)
```

Supported files

Most files are read by skipping line 1 (title) and line 3 (units) and taking line 2 as the column names. `crop_yld_aa.txt` and `crop_yld_yr.txt` skip two lines. `lu_change_out.txt`, `basin_crop_yld_aa.txt` and `basin_crop_yld_yr.txt` keep line 3 as data. `checker.out` and `files_out.out` are read with fixed column names.

```
files_out.out  checker.out  hru_orgc.txt
hru_wb_aa.txt  hru_ncycle_aa.txt  hru_nb_aa.txt  hru_carbon_aa.txt  hru_nut_carb_gl_aa.txt  hru_ls_aa.txt
basin_wb_aa.txt  basin_nb_aa.txt  basin_carbon_aa.txt  basin_ls_aa.txt  basin_pw_aa.txt
crop_yld_yr.txt  crop_yld_aa.txt  lu_change_out.txt  basin_crop_yld_yr.txt  basin_crop_yld_aa.txt
hydout_aa.txt  hydin_aa.txt  deposition_aa.txt  wetland_aa.txt  basin_aqu_aa.txt  basin_res_aa.txt
recall_aa.txt  basin_cha_aa.txt  basin_sd_cha_aa.txt  basin_sd_chamorph_aa.txt  basin_psc_aa.txt  ru_aa.tx
```

Only average annual files are covered. `hru_orgc.txt` and `hru_nut_carb_gl_aa.txt` are not written by 62.0.1.

Limits with 62.0.1

- The script stops (exit code 1) at the first file listed in `files_out.out` that it has no instructions for, with the message `Swat+ output filename not found in read specification dictionary.` A 62.0.1 run usually lists such files (for example the carbon files, `sd_chanbud_aa.txt` or any daily file), so a full run stops part way. Reading single files with `read_output` works.
- It passes `delim_whitespace=True` to `pandas.read_csv`. pandas 2.2 warns about this option and pandas 3.0 no longer accepts it; with pandas 3.0 every file fails with the message `could not be parsed for some reason`.
- `basin_wb_aa.txt` ends with the two-word label `Original Simulation` and a number, which gives its data row more values than the header has names, so pandas moves the leading columns into the row index.
- `files_out.out` is read as two columns, tag and file name. With `csvout` on, some tags contain a space (`CHANBUD ORDER`), which breaks those lines.

Related files

- Output naming convention: output file names and layout.
- Checker and diagnostics: `files_out.out` and `checker.out`.
- Water balance outputs: the trailing label in `basin_wb_aa.txt`.

Sources

- `test/spcheck.py`
- `test/swat_output_to_csv_files.py`
- Regression test: `CMakeLists.txt` and `refdata/Ames_sub1/.testfiles.txt`
- `files_out.out`: `proc_bsn.f90`; removed carbon files noted in `output_landscape_init.f90`